

DEPARTMENT OF DATA SCIENCE AND ARTIFICIAL INTELLIGENCE
INDIAN INSTITUTE OF TECHNOLOGY MADRAS
CHENNAI – 600036

Anomaly Detection Methods

A Report

Submitted by

HOBHAV S FULZELE

For the award of the degree

Of

MASTER OF TECHNOLOGY

May 2026



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The art of detection lies in understanding
what is normal to find what is abnormal.

– Anonymous

THESIS CERTIFICATE

This is to undertake that the Report titled **ANOMALY DETECTION METHODS**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master of Technology**, is a bona fide record of the research work done by me under the supervision of **Dr. Sivaram Ambikasaran and Dr. Sri Vallabha Deevi**. The contents of this Report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Anomaly detection is essential in applications where unusual patterns may indicate faults, security threats, fraudulent behaviour, or medical emergencies. This report presents a comparative evaluation of classical machine learning and deep learning based anomaly detection methods across a wide range of data modalities, including uni-variate and multivariate time series, tabular transactional data, textual data, audio signals, surveillance video, synthetic media, and high-dimensional medical and industrial images. Multiple benchmark datasets were considered, including MIT-BIH ECG signals for cardiac abnormalities, NAB temperature data for temporal anomalies, smart-home activity records for elderly fall detection, financial transactions for fraud detection, AITEX textile fabric images for defect identification, brain magnetic resonance imaging scans for tumor-related abnormalities, and additional datasets for offensive language detection, sentiment analysis, industrial machine faults, spoofed speech, deepfake media, and AI-generated text detection.

The experiments incorporate statistical techniques, distance-based and isolation-based approaches, clustering methods, probabilistic models, and autoencoder architectures with preprocessing tailored to each domain. Performance was assessed using standard anomaly detection metrics to enable consistent comparison across datasets. Overall, the findings highlight that anomaly detection is strongly data-dependent, and selecting an appropriate method requires careful consideration of the dataset's structure, modality, feature representation, and the nature of anomalies present in the data.

KEYWORDS Anomaly Detection; Uni-variate data; Multi-variate data; High-dimensional data; Medical images; Benchmark datasets; Defect identification.

Code Sourse Link:

https://github.com/HSF0105/Evaluation-of-Anomaly-Detection-Methods

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GLOSSARY

Anomaly	An observation that deviates significantly from normal patterns in data, often indicating rare or abnormal events.
Autoencoder	A neural network used to learn compressed representations of data and reconstruct inputs. In anomaly detection, anomalies are identified through high reconstruction error.
Gaussian Mixture Model (GMM)	A probabilistic model that represents data as a mixture of Gaussian distributions; low likelihood samples are treated as anomalies.
Principal Component Analysis (PCA)	A dimensionality reduction method that identifies anomalies using projection residuals or reconstruction errors.
Reconstruction Error	The difference between an input and its reconstructed output (generally from a model), used as an anomaly score.

ABBREVIATIONS

DBSCAN Density-Based Spatial Clustering of Applications with Noise.

ECG Electrocardiogram.

LOF Local Outlier Factor.

MCD Minimum Covariance Determinant.

MIT-BIH MIT-BIH Arrhythmia Database.

MRI Magnetic Resonance Imaging.

NAB Numenta Anomaly Benchmark Dataset.

OC-SVM One-Class Support Vector Machine.

SIFT Scale-Invariant Feature Transform.

SURF Speeded Up Robust Features.

SVM Support Vector Machine.

TP, FP, FN True Positive, False Positive, False Negative (classification metrics).

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Anomaly detection refers to the identification of data patterns, events, or observations that deviate significantly from expected behavior. These deviations are often rare by nature and can indicate critical conditions such as system faults, security breaches, fraudulent activities, medical abnormalities, or safety risks. As modern systems increasingly operate in real time and at large scale, robust anomaly detection has become essential across domains, such as healthcare, finance, smart homes, manufacturing, and industrial monitoring.

However, anomalies manifest differently depending on the underlying data modality. In time series data, anomalies may appear as sudden spikes, drops, or unusual temporal patterns. In tabular transactional data, anomalies often indicate fraudulent or malicious actions hidden among large volumes of normal activity. In images, anomalies typically represent spatial irregularities, structural defects, or pathological regions. Each domain presents unique challenges, making anomaly detection a fundamentally data-dependent task rather than a single unified problem.

1.2 PROBLEM STATEMENT

Although numerous anomaly detection algorithms exist, no single approach performs best across all types of data. Key challenges include:

- **Variability in data structure:** univariate vs multivariate time series, structured tabular data, medical or industrial images and sentimental deviation or Rare/Infrequent terms in texts.
- **Different forms of anomalies:** point, contextual, and collective.

- **Different noise levels, imbalance ratios, and feature distributions.**
- **Need for methods that can handle rare events**, limited labeled data, and domain-specific constraints.

Therefore, the core problem addressed in this report is:

To systematically evaluate classical and deep learning anomaly detection techniques across multiple data modalities and determine how well these methods adapt to different types of anomalies and datasets.

1.3 OBJECTIVES

To address the problem statement, this report aims to achieve the following objectives:

1. **Analyze and preprocess datasets** from different domains:
    - MIT-BIH ECG signals, NAB temperature anomalies, Smart-home elderly people fall detection events
    - Financial fraud transactions, Textile fabric defect images, Brain MRI scans for tumor anomalies
    - Offensive language identification, large movie reviews, Anomalous pedestrian motion patterns in videos
    - Surveillance videos for realistic anomalies, Malfunctioning machine inspection using sound, Spoofed and synthetic speech detection audio data
    - Machine generated text detection, Machine generated video or deep fake detection
2. **Implement a variety of anomaly detection methods**, including Statistical, distance-based techniques and Isolation Forest, One-Class SVM, and Autoencoders
3. **Develop domain-specific pipelines** tailored to each dataset's properties.
4. **Evaluate and compare methods** using appropriate metrics such as precision, recall, F1-score, reconstruction error.
5. **Identify key insights** on which methods perform best for which data types.

CHAPTER 2

LITERATURE REVIEW

Anomaly detection has been studied across several decades and spans multiple disciplines, including statistics, data mining, machine learning, signal processing, image analysis, and deep learning. This chapter reviews foundational concepts, classical algorithms, deep learning approaches, and domain-specific literature relevant to the datasets and modalities used in this report.

2.1 OVERVIEW OF ANOMALY DETECTION

Anomalies, also called outliers, are observations that deviate significantly from expected patterns. The literature commonly categorizes anomalies into (Chandola et al. (2009)[9]):

1. **Point anomalies:** Individual data points that are unusual, like outliers.
2. **Contextual anomalies:** Anomalous only within a specific context (e.g., seasonal patterns).
3. **Collective anomalies:** A sequence or group of points that deviate collectively, even if individual values appear normal.

Various methods have been proposed based on assumptions about data distribution, label availability, and computational constraints. Broadly, the methods fall into the categories of *statistical*, *distance-based*, *machine learning*, and *deep learning*.

2.2 CLASSICAL STATISTICAL AND DISTANCE-BASED METHODS

2.2.1 Z-Score and Statistical Thresholding

One of the earliest approaches involves estimating the mean and standard deviation of data and flagging points outside a threshold. While simple, these models fail when the data is non-Gaussian, contains trends, or exhibits seasonality.

2.2.2 Moving Average, Forecasting Residuals, and Control Charts

Techniques such as moving averages, exponentially weighted moving averages (EWMA), and Holt–Winters forecasting are widely used in industry for temporal anomaly detection. An observation is considered anomalous when forecasting residuals (i.e., the difference between predicted and actual values) exceed expected limits.

2.2.3 Distance-Based Methods

Distance-based methods assume anomalies lie far from dense clusters (Chandola et al. (2009)[9]). Key approaches are:

- **k-Nearest Neighbors (k-NN):** Computes distances to neighbors; large distances signal anomalies.
- **Local Outlier Factor (LOF):** Measures local deviation of density; lower density points are considered anomalous.

These methods struggle with high-dimensional data due to the curse of dimensionality.

2.3 STATISTICAL MODELING AND PROBABILISTIC APPROACHES

2.3.1 Elliptic Envelope

The Elliptic Envelope algorithm is an unsupervised machine learning method that uses covariance estimation on Gaussian distribution data (Rousseeuw (1984)[24]). Elliptic envelope tries to make an elliptical cluster and fits the major class instances in that. Instances far away from the cluster are then considered as anomalies (Ashrafuzzaman et al. (2020)[4]). The Elliptic Envelope method assumes a single Gaussian distribution and relies on covariance-based distance measures; therefore, its performance deteriorates in high-dimensional spaces, non-Gaussian data distributions, and datasets with multiple clusters where a single elliptical boundary is insufficient (Ashrafuzzaman et al. (2020)[4]).

2.3.2 Principal Component Analysis (PCA) for Anomaly Detection

In PCA-based anomaly detection, the residual vector represents the difference between the original observation and its reconstruction using principal components. The squared

magnitude of this residual, known as the Q-statistic (or Squared Prediction Error), is used to detect anomalies, where large values indicate significant deviations from the model (Jackson and Mudholkar (1979)[15]). Despite its simplicity, PCA is sensitive to nonlinear patterns, motivating extensions such as Kernel PCA and autoencoder-based methods.

2.3.3 Gaussian Mixture Models

Gaussian Mixture Models (GMMs) represent data as a combination of multiple Gaussian distributions, each capturing a cluster or pattern (Bishop (2006)[5]). They learn the means, covariances, and weights using the EM algorithm. For anomaly detection, GMMs evaluate how likely each data point is under the model—points with very low likelihood are flagged as anomalies. This approach works well when normal data forms clear clusters. GMMs are interpretable and effective for multi-cluster data, but they perform poorly when data is highly non-Gaussian or high-dimensional.

2.4 MACHINE LEARNING-BASED APPROACHES

2.4.1 Isolation Forest

Isolation Forest isolates anomalies by randomly partitioning feature space (Liu et al. (2008)[19]). Anomalies require fewer splits, making the method efficient even for large, high-dimensional datasets. It has been widely used for financial fraud and tabular anomalies.

2.4.2 One-Class SVM

One-Class SVM learns a decision boundary around normal data. Points falling outside the boundary are classified as anomalies (Chandola et al. (2009)[9]). It works well for medium-sized datasets but becomes computationally expensive for large or high-dimensional data.

2.4.3 Clustering-Based Approaches

Clustering algorithms such as k-Means and DBSCAN are also used for anomaly detection, where points that lie far from cluster centroids or those labeled as noise by DBSCAN are considered anomalous (Chandola et al. (2009)[9]). However, these methods rely on the assumption that the data naturally forms meaningful clusters, an assumption that may not hold for more complex datasets.

2.4.4 AdaBoost for Anomaly Detection

AdaBoost is an ensemble learning algorithm that builds a strong classifier by combining multiple weak learners, typically shallow decision trees (Freund and Schapire (1995)[12]). Although originally designed for supervised classification, AdaBoost has been used in anomaly detection by training it on datasets where anomalies are treated as a minority class or by interpreting low-confidence predictions as outliers. The method benefits from being fast, interpretable, and well-suited to tabular data. However, its performance is highly dependent on the quality and balance of labeled data, which can be challenging in anomaly detection settings where anomalies are rare and often underrepresented.

2.4.5 Autoencoders

Autoencoders are the primary deep learning method used in this report for anomaly detection. They are neural networks trained to compress input data into a lower-dimensional representation and then reconstruct it. When trained only on normal samples, the network learns the typical structure and patterns present in the data. As a result, it reconstructs normal inputs accurately but struggles to reconstruct abnormal or unseen patterns. Anomalies are detected by computing the reconstruction error between the input and its reconstructed output (Chalapathy and Chawla (2019)[8]).

- Low reconstruction error $\rightarrow$ normal sample
- High reconstruction error $\rightarrow$ potential anomaly

A suitable threshold is selected to classify anomalies based on this error distribution.

CHAPTER 3

DATASET DESCRIPTION

3.1 MIT-BIH ARRHYTHMIA DATABASE

The MIT-BIH Arrhythmia Database (Goldberger et al. (2000)[14]) contains 48 half-hour, two-channel ECG recordings from 47 subjects, digitized at 360 Hz with 11-bit resolution. Of these, 23 recordings were randomly selected, while the remaining 25 include rarer but clinically important arrhythmias. It is widely used as a benchmark dataset for arrhythmia detection and heartbeat abnormality research. In this work a cleaned version of the dataset available on Kaggle (Mondejar (2024)[22]) was used.

Several other similar datasets are also widely available, including the PTB Diagnostic ECG Database (PhysioNet), the MIT-BIH Normal Sinus Rhythm Database, and the Chapman University Arrhythmia Dataset.

3.2 TEMPERATURE (NAB)

The Numenta Anomaly Benchmark (NAB) (Ahmad et al. (2017)[1]) is a real-time time-series anomaly benchmark composed of 58 datasets from domains such as environmental data, server metrics, and sensor logs. It contains over 365,000 data points with labeled anomalies, making it ideal for evaluating temporal anomaly detection models. In this work a cleaned version of the dataset available on Kaggle (boltzmannbrain (2017)[6]) was used.

Several other similar datasets are also widely available, including the Yahoo S5 Time-Series Anomaly Dataset, the UCR Time-Series Anomaly Archive Database, and the SMD (Server Machine Dataset).

3.3 SMART-HOME FALL DETECTION DATASET

The Anomaly Detection – Falling People dataset (Vidulin and Krivec (2010)[30]) is a time-series sensor dataset collected from a smart-home environment for fall detection. It consists of 25 CSV files (20 for training and 5 for testing), where each file represents data from an individual subject.

Each record contains multivariate sensor readings with 3D spatial features (X, Y, Z) along with one-hot encoded activity signals from 4 body-mounted sensors (chest, ankles, and belt).In this work a cleaned version of the dataset available on Kaggle (jorekai (2021)[17]) was used.

Several other similar datasets are also widely available, including the SisFall Dataset (accelerometer/gyroscope-based falls), the UMAFall Dataset, and the UR Fall Detection Dataset.

3.4 FINANCIAL FRAUD TRANSACTION DATASET

The Metaverse Financial Transactions Dataset (Janjua (2024)[16]) contains anonymized transaction records used for fraud detection studies. It includes details such as transaction amount, time, type, device, location, and flags indicating legitimate or fraudulent behavior. The dataset is highly imbalanced, reflecting realistic rare-event fraud patterns.

Several other similar datasets are also widely available, including the IEEE-CIS Fraud Detection Dataset, the Credit Card Fraud Dataset (ULB), and the PaySim Mobile Money Fraud Dataset.

3.5 TEXTILE FABRIC DEFECT DATASET (AITEX) (IMAGE)

The AITEX Fabric Database (Silvestre-Blanes et al. (2019)[26]) contains high-resolution textile images representing seven fabric types with and without defects. Image sizes are typically large (up to 4096 × 256), and the dataset includes defective and defect-free samples. The dataset is widely used in industrial anomaly detection research for surface inspection.

Several other similar datasets are also widely available, including the DAGM 2007 Surface Defect Dataset, the MVTec Anomaly Detection Dataset, and the Severstal Steel Defect Dataset.



(a) Defective textile fabric image



(b) Ground truth mask of the defect



(c) Non-defective textile fabric image

Figure 3.1: Examples of textile fabric inspection dataset showing defective and non-defective samples along with corresponding ground truth mask.

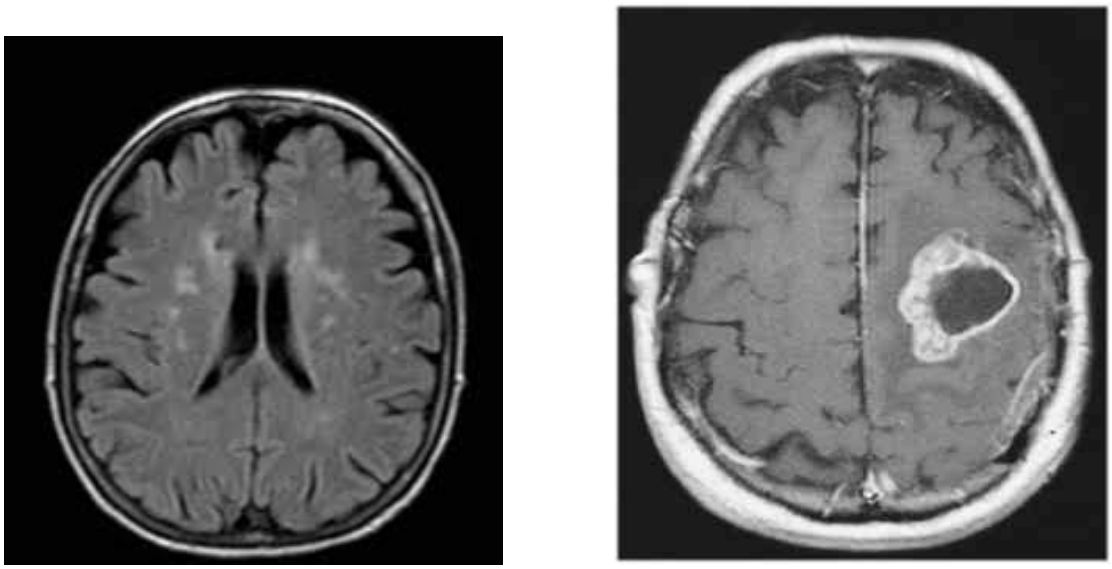
3.6 BRAIN MRI ANOMALY DATASET (IMAGE)

The Brain MRI Images dataset (Chakrabarty (2019)[7]) contains normal and tumor MRI scans from multiple patients. Images include axial brain slices labeled as either tumor-present or tumor-free. This dataset is widely used for medical anomaly detection, especially tumor identification.

Several other similar datasets are also widely available, including the BraTS (Brain Tumor Segmentation Challenge), the Figshare Brain MRI Dataset, and the RIDER NEURO MRI Dataset.

3.7 OFFENSIVE LANGUAGE IDENTIFICATION DATASET (TEXT)

The Offensive Language Identification Dataset (Zampieri et al. (2019)[32]) is a text classification dataset consisting of approximately 14,000 annotated tweets. Each instance is labeled hierarchically into offensive (OFF) or non-offensive (NOT), with further subcategories identifying targeted insults (TIN) and untargeted profanity (UNT).



(a) Normal brain MRI (no tumor)

(b) Brain MRI with tumor

Figure 3.2: Comparison of brain MRI scans showing a normal case and a tumor-affected case.

The dataset contains short user-generated texts with categorical labels, making it suitable for multi-level classification tasks. It is moderately imbalanced, with offensive content forming a smaller portion of the data, reflecting real-world distributions of toxic language. Several other similar datasets are also widely available, including the HateXplain Dataset, the Davidson Hate Speech Dataset, and the Jigsaw Toxic Comment Dataset.

3.8 IMDB LARGE MOVIE REVIEW DATASET (TEXT)

The IMDB Large Movie Review Dataset (Maas et al. (2011)[20]) is a text classification dataset containing 50,000 movie reviews, evenly split into 25,000 training and 25,000 testing samples. Each review is labeled as positive or negative sentiment, resulting in a balanced class distribution (50% each).

The dataset consists of long-form user-generated text, making it suitable for evaluating NLP models. Its size and balanced nature support sentiment classification, while also enabling anomaly detection tasks such as identifying outlier reviews, spam, or inconsistent sentiment patterns.

Several other similar datasets are also widely available, including the Amazon Reviews Dataset, the Yelp Review Polarity Dataset, and the Rotten Tomatoes Reviews Dataset.

3.9 UCSD ANOMALY DETECTION DATASET (VIDEO)

The UCSD Pedestrian Dataset is a benchmark dataset for video anomaly detection, featuring surveillance footage of pedestrian walkways. Normal activity involves pedestrian movement, while anomalies include bicycles, wheelchairs, carts, skaters, and other unusual motion patterns. It is widely used to evaluate spatio-temporal anomaly detection methods in computer vision (Mahadevan et al. (2013)[21]).

Several other similar datasets are also widely available, including the Avenue Dataset, the ShanghaiTech Campus Dataset, and the Subway Entrance Dataset.



(a) Normal Image (b) Mask of anomalous Image (c) Anomalous Image

Figure 3.3: Examples of images from the UCSD dataset illustrating anomaly conditions with corresponding mask and a normal condition image.

3.10 UCF CRIME DATASET (VIDEO)

The UCF Crime Dataset is a large-scale real-world surveillance anomaly dataset containing long untrimmed videos with abnormal events such as robbery, burglary, assault, road accidents, fire, fighting, and vandalism. Normal videos contain regular daily activities (Sultani et al. (2018)[29]).

Several other similar datasets are also widely available, including the XD-Violence Dataset, the ShanghaiTech Dataset, and the Avenue Dataset.



(a) Normal activity (office environment)



(b) Anomalous activity (violent event)

Figure 3.4: Sample frames from the UCF Crime dataset illustrating normal and anomalous activities captured in surveillance videos.

3.11 MIMII DATASET (AUDIO)

The MIMII (Malfunctioning Industrial Machine Investigation and Inspection) Dataset contains machine sounds recorded from valves, pumps, fans, and slide rails under both normal and faulty operating conditions. It is a benchmark dataset for unsupervised industrial acoustic anomaly detection (Purohit et al. (2019)[23]).

Several other similar datasets are also widely available, including the ToyADMOS Dataset and the DCASE Machine Sound Dataset.

3.12 ASVSPOOF 2019 DATASET (AUDIO)

The ASVspoof 2019 Dataset is designed for detecting spoofed and synthesized speech. It contains genuine speech samples and attacks generated through replay, voice conversion, and text-to-speech systems. It is widely used in speech security and synthetic voice anomaly detection (Yamagishi et al. (2019)[31]).

Several other similar datasets are also widely available, including the WaveFake Dataset and the FakeAVCeleb Dataset.

3.13 RAID DATASET (TEXT)

The RAID Dataset is a large-scale benchmark designed for distinguishing AI-generated and human-written text. It contains outputs from multiple language models alongside authentic human-authored documents across varied prompts and writing styles. It is useful for benchmarking machine-generated text detection systems (Dugan et al. (2024)[11]).

Several other similar datasets are also widely available, including the HC3 Dataset, Ghostbuster Benchmark, and the DetectGPT Evaluation Corpora.

3.14 AI VS HUMAN TEXT

This dataset was released on focused on identifying essays written by AI systems versus human. It contains data from multiple sources which where add together to form this dataset (Gerami (2024)[13]).

Several other similar datasets are also widely available, including the M-DAIGT Dataset and HC3 Dataset.

3.15 WILDDEEPFAKE DATASET

WildDeepfake is a real-world deepfake face dataset collected from internet sources containing manipulated and authentic face videos/images under unconstrained conditions such as varied lighting, pose, blur, and compression. It is more challenging than laboratory datasets (Zi et al. (2020)[33]).

Several other similar datasets are also widely available, including the FaceForensics++ Dataset and DF-Platter Dataset.

3.16 CELEB-DF (V2)

Celeb-DF (v2) is a high-quality deepfake video dataset containing realistic face-swapped celebrity videos with fewer visual artifacts than earlier datasets. It is widely used for evaluating generalizable deepfake detection systems (Li et al. (2020)[18]; Suju (2023)[28]).

Several other similar datasets are also widely available, including the FaceForensics++ Dataset and the DFDC Dataset.

3.17 DATASETS USED FOR ADDITIONAL EXPERIMENTS IN MACHINE GENERATED TEXT DETECTION

1. **M-DAIGT Dataset**: MDAIGT (Multi-Domain AI-Generated Text Detection) is a dataset designed to support research on detecting AI-generated English text across multiple domains, including news and academic writing. The dataset includes both human-written and machine-generated samples, aiming to help evaluate the robustness and domain generalization of AI-text detection models (Cognitive Software Automation and Language Research Group (CogniSAL) (2024)[10]).

2. **CT2 Dataset**: The data will be structured such that each prompt will have a human-written story and corresponding parallel generations from all the included LLMs. This dataset offers a diverse set of AI-generated content for participants to analyze and classify (Roy et al. (2025)[25]; Singh et al. (2025)[27]).

3. **AIGTxt Dataset**: The AI-Generated Text (AIGTxt) dataset, which was used in the research project Deep Learning Detection Method for Large Language Models-Generated Scientific Content. The dataset was collected from published academic articles prior to 2022 and covers a wide range of fields. Specifically, the dataset includes scientific publications from 10 distinct fields of study (Alhijawi et al. (2024b)[3]; Alhijawi et al. (2024a)[2]).

CHAPTER 4

METHODOLOGY

This chapter describes the methodological framework adopted for anomaly detection across the ECG, temperature, smart-home fall, financial fraud, textile fabric, and brain MRI datasets. The experiments employed a consistent pipeline combining classical machine-learning anomaly detection techniques with autoencoder-based reconstruction analysis. Each method was applied uniformly across datasets to enable domain-independent comparison.

4.1 ANOMALY DETECTION FRAMEWORK

Given a dataset

$$X = \{x_1, x_2, \ldots, x_n\}, \qquad x_i \in \mathbb{R}^d, \tag{4.1}$$

Each method computes an anomaly score

$$s(x_i) \in \mathbb{R},$$

Anomalies are detected using a threshold. That is,

$$x_i \text{ is anomaly if } s(x_i) > \tau,$$

where τ is typically chosen as a high percentile of the score distribution (e.g., 95th percentile).

All models ultimately produce binary labels:

$$y_i = \begin{cases} 1, & \text{normal,} \\ -1, & \text{anomaly.} \end{cases}$$

4.2 AUTOENCODER-BASED ANOMALY DETECTION

Autoencoders constitute the primary deep learning component of the methodology. They are trained exclusively on normal samples to learn compact representations of normal behaviour. During inference, samples exhibiting unusually high reconstruction error are classified as anomalies.

4.2.1 Architecture and Training Strategy

Across all datasets, the autoencoders follow a consistent design. Each model uses an encoder–decoder architecture, where fully connected layers are applied for tabular or time-series inputs and convolutional layers are used for image-based data. The models are trained using mean squared reconstruction error as the loss function and the Adam optimizer with its default parameters. During training, only normal samples are included so that the network learns the underlying patterns of normal behavior.

4.2.2 Anomaly Scoring

After reconstruction, an anomaly score is computed using:

$$\text{Reconstruction Error} = \|x - \hat{x}\|_2^2 \tag{4.2}$$

A threshold (often based on a high reconstruction-error percentile) separates normal and anomalous observations.

4.3 EVALUATION PROCEDURE

All models are evaluated using the same metrics to ensure fairness across datasets. The evaluation includes:

- **Precision** – proportion of detected anomalies that are correct.
- **Recall** – proportion of true anomalies successfully identified.
- **F1-Score** - harmonic mean of precision and recall.

Outputs from each classical and autoencoder model are compiled into a unified results table that contains method names, metric values, and supervision type.

4.4 METHODOLOGICAL CONSISTENCY ACROSS DATASETS

Although the datasets differ considerably in format, including time-series, tabular, image, and activity-event data, the methodology remains consistent across all experiments. This consistency is maintained through uniform feature scaling, the use of the same anomaly-detection models, reconstruction-based scoring for deep-learning approaches, standardized evaluation metrics, and percentile-based thresholds for classifying anomalies. Together, these elements provide a unified framework that enables objective comparison between classical machine-learning techniques and autoencoder-based methods across different anomaly-detection scenarios.

CHAPTER 5

EXPERIMENTAL SETUP

This chapter presents the computational environment, model configurations, and evaluation procedures followed in all anomaly detection experiments. The setup was kept consistent across the ECG, temperature, fall-detection, financial fraud, textile fabric, and brain MRI datasets to ensure comparability of results.

5.1 COMPUTATIONAL ENVIRONMENT

All experiments were conducted using a Python-based machine learning environment. The following libraries formed the core of the experimental pipeline:

- **NumPy** and **Pandas** for numerical processing
- **Scikit-learn** for classical anomaly detection algorithms, feature scaling, PCA, and evaluation
- **PyTorch** for constructing and training autoencoders
- **Matplotlib/Seaborn** for visualization
- **OpenCV** for reading fabric and MRI images

The experiments were executed on a standard CPU runtime. No GPU-specific implementation or CUDA-accelerated operations were used.

5.2 DATA PREPARATION AND FEATURE EXTRACTION

Textile Fabric Defect Dataset (AITEX)

- Image normalization: Pixel values normalized after reading images in grayscale.
- Resizing: Images were resized to a consistent shape (256×256) prior to model input.

- Feature exploration: Local Binary Pattern (LBP)–based texture features were explored.

Brain MRI Anomaly Dataset

- **Unified size and normalization**: All MRI images were converted to grayscale and normalized for stable autoencoder training.
- **Automated dataset preparation**: The custom pipeline method handled loading, partitioning, and label arrangement.

Offensive Language Identification Dataset

For this study, three different feature extraction methods were prepared and evaluated. One representation was created by removing records with missing text and converting the original labels into a binary anomaly-detection setting, where offensive posts were treated as anomalies and non-offensive posts as normal samples. Because the dataset consists of short, noisy, and socially contextual text, multiple feature spaces were explored across experiments.

The first representation employed lexical TF–IDF features along with simple surface-level indicators such as text length, word count, uppercase character usage, and punctuation counts. The second representation utilized dense sentence embeddings obtained from the `all-MiniLM-L6-v2` model. The third, more comprehensive classical NLP pipeline combined bag-of-words vectors, word-level TF–IDF, character n-grams, part-of-speech counts, heuristic offensive-word indicators, stylistic features, and averaged Word2Vec embeddings.

After feature extraction, the resulting vectors were standardized prior to anomaly scoring.

IMDB Large Movie Review Dataset

To cast IMDB sentiment analysis into an anomaly-detection setting, the binary labels were converted so that negative reviews represented anomalies and positive reviews represented the normal class. Since the reviews are substantially longer than tweets, a broader set of textual descriptors was extracted, including bag-of-words features, TF–IDF

unigrams and bigrams, character n-grams, part-of-speech statistics, lexical and stylistic measures, sentiment-related attributes, and averaged Word2Vec document embeddings. These complementary sparse and dense features were concatenated to obtain a richer representation of each review, and the final vectors were standardized before model training.

UCSD Anomaly Detection Dataset

The UCSD Pedestrian Dataset videos were processed at the frame level using the official train–test split, with pixel-level ground-truth masks used only to derive frame-level labels for evaluation.

Five different feature extraction methods were prepared and evaluated to compare appearance-based and motion-aware feature spaces. The main pipeline extracted deep frame embeddings from RGB frames using a pretrained **ResNet** backbone. Additional experiments employed handcrafted computer-vision descriptors, including **Histogram of Oriented Gradients (HOG)** computed from resized grayscale frames and dense **optical-flow features** computed between consecutive frames to capture motion irregularities.

Vision Transformer embeddings were also extracted from resized and normalized RGB frames, and a **memory-augmented autoencoder** variant was evaluated on grayscale frame inputs. All feature vectors were standardized prior to applying the anomaly-detection models.

UCF Crime Dataset

The UCF-Crime videos were transformed into fixed-length clips by uniformly sampling frames from each video, with normal videos used to model the normal class and anomalous videos reserved for evaluation. In the primary spatio-temporal pipeline, 32 frames per video were resized to 112×112 pixels, normalized, and stacked into short clips from which a frozen 3D convolutional network extracted compact clip-level

descriptors. An alternative transformer-based pipeline sampled 16 frames per video, resized them to 224×224, applied standard image normalization and light augmentation during training, and extracted Vision Transformer frame embeddings that were then temporally aggregated into a single video-level representation. The resulting feature vectors were standardized before anomaly detection so that all classical and deep detectors operated on comparable input scales.

MIMII Dataset

The MIMII machine-sound recordings were converted into fixed-duration audio inputs so that all files could be represented with a common feature length. In the primary handcrafted pipeline, the signals were resampled and summarized using log-Mel spectrogram bands, MFCC coefficients, delta-MFCC features, and complementary spectral descriptors such as spectral centroid, bandwidth, roll-off, zero-crossing rate, and RMS energy. These time-varying features were aggregated using summary statistics to produce a compact representation suitable for classical anomaly detectors. In a second representation pipeline, each waveform was resampled to a common sampling rate, truncated or zero-padded to a fixed duration, and passed through a pretrained speech encoder such as HuBERT or wav2vec 2.0; mean pooling of the hidden states yielded a dense utterance-level embedding. All audio feature vectors were standardized before anomaly scoring.

ASVspoof 2019 Dataset

For ASVspoof 2019, genuine speech was treated as the normal class and spoofed utterances were treated as anomalies. The primary acoustic pipeline applied pre-emphasis and then extracted log-Mel spectrograms, MFCCs, and first- and second-order MFCC derivatives from each utterance. To convert variable-length speech signals into a fixed-size representation, each coefficient trajectory was summarized through statistical pooling using measures such as mean, standard deviation, skewness, kurtosis, minimum,

maximum, and median. In the alternative embedding-based pipeline, waveforms were resampled to 16 kHz, truncated or padded to a common duration, and encoded with a pretrained wav2vec 2.0 model, after which the hidden states were mean-pooled to obtain utterance-level embeddings. The resulting features were standardized before model fitting.

RAID Dataset

The RAID corpus was converted into a binary anomaly-detection task by treating machine-generated documents as anomalies and human-written documents as the normal class. To compare sparse, linguistic, and dense semantic representations, three preprocessing pipelines were evaluated. The classical pipeline used TF–IDF unigram and bigram features, reduced the sparse representation using Truncated SVD, and concatenated the reduced vectors with handcrafted linguistic indicators such as document length, punctuation frequencies, lexical diversity, paragraph-structure variation, sentiment polarity and subjectivity, repeated-token patterns, and counts of AI-associated cue phrases. A second pipeline generated transformer embeddings using a pretrained DistilBERT model with mean pooling across tokens. A third dense representation used sentence-transformer embeddings based on `all-MiniLM-L6-v2`. Each document-level representation was standardized before anomaly detection.

AI Vs Human Text

The Kaggle AI vs Human Essays Dataset essays were first converted into a common text-and-label format, with AI-generated essays treated as anomalies and human-written essays as normal samples. As with RAID, multiple feature spaces were prepared to compare classical and modern text representations.

The classical pipeline combined TF-IDF unigram and bigram features with Truncated SVD, along with a set of handcrafted linguistic attributes capturing document length,

punctuation usage, lexical diversity, paragraph-level variation, sentiment cues, repeated-word patterns, and characteristic AI-style phrases. A transformer-based pipeline then represented each essay using mean-pooled DistilBERT embeddings, while a third pipeline employed dense sentence embeddings derived from the `all-MiniLM-L6-v2` model.

All resulting feature vectors were standardized prior to applying the anomaly-detection models.

WildDeepfake Dataset

The WildDeepfake data were organized by real and fake identity folders, and a fixed number of face images were sampled uniformly from each identity in order to reduce redundancy while preserving intra-subject variation. Three alternative visual representations were prepared. The first used a pretrained EfficientNet-B0 backbone to extract image-level features from RGB inputs resized to 224×224 pixels. The second arranged 16 sampled images from the same identity into pseudo-temporal clips and fed them to a lightweight 3D convolutional network after resizing each frame to 112×112, thereby enabling the model to capture short-range spatial correlations across the sampled sequence. The third used Vision Transformer embeddings extracted from resized and normalized 224×224 images. Where multiple images were sampled for one identity, the image-level descriptors were aggregated into a single representation before anomaly detection, and all resulting features were standardized.

Celeb-DF (v2)

Celeb-DF videos were converted into fixed-size video-level representations by uniformly sampling frames from each clip and applying consistent spatial normalization. In the CNN-based pipeline, sampled RGB frames were resized to 224×224 and passed through a pretrained EfficientNet-B0 backbone; the resulting frame descriptors were mean-pooled to obtain a single feature vector for each video. In the spatio-temporal pipeline, 16

frames per video were resized to 112×112 and grouped into short clips that were processed by a lightweight 3D convolutional network to capture temporal manipulation cues. A third representation used a Vision Transformer, where the sampled frames were resized to 224×224, normalized using ImageNet statistics, and converted to CLS-token embeddings that were aggregated across frames. The final video-level descriptors were standardized before anomaly scoring.

Datasets Used For Additional Experiments in Machine Generated Text Detection

For this pooled machine-generated-text benchmark, the datasets were not evaluated separately; instead, texts from multiple corpora were harmonized and merged into one common benchmark. The combined collection included the AI-vs-Human essay dataset, RAID, and, in the broader pooled setting, additional machine-generated text sources such as M-DAIGT, CT2, and AIGTxt. Each source was converted into a common two-column schema consisting of raw text and a binary label, where human-written text represented the normal class and machine-generated text represented anomalies. During preprocessing, records with missing text were removed, label conventions were standardized across datasets, whitespace and formatting inconsistencies were cleaned, and extremely short or unusable samples were discarded. After normalization, the corpora were concatenated, shuffled, optionally down-sampled or balanced by class to control memory usage and reduce source dominance, and finally split under a unified train-test evaluation protocol.

To compare how representation choice affects cross-dataset generalization, the merged text collection was then evaluated under multiple feature pipelines. A classical mixed-feature baseline used sparse lexical representations and handcrafted linguistic indicators, a transformer-based pipeline used mean-pooled DistilBERT embeddings, and a dense semantic pipeline used sentence embeddings from `all-MiniLM-L6-v2`. The resulting document-level feature vectors were standardized before anomaly detection. In this way,

this work measures whether methods that work well on individual machine-generated text datasets such as RAID and AI-vs-Human text remain effective when all machine-generated text sources are combined and evaluated under one common protocol.

5.3 EVALUATION METRICS

The performance of all the methods mentioned in Chapter 4 was evaluated using standard binary classification metrics:

$$\text{Precision} = \frac{TP}{TP+FP} \quad \text{Recall} = \frac{TP}{TP+FN}$$
$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision}+\text{Recall}} \tag{5.1}$$

These metrics quantify the balance between detection accuracy and false alarms across methods and datasets. All the metrics were calculated only for anomalous datapoints, as we are interested in them.

CHAPTER 6

RESULTS

This chapter presents the anomaly detection results obtained across all six datasets using classical machine-learning methods, PCA-based reconstruction scoring, Gaussian mixture modelling, and autoencoder-based reconstruction error. For each dataset, Precision, Recall, and F1-Score are reported.

6.1 MIT-BIH ARRHYTHMIA DATASET

AdaBoost attains the highest precision (0.923), followed by PCA (0.576) and OC-SVM (0.546), showing AdaBoost produces the fewest false alarms on ECG.

DBSCAN reaches perfect recall (1.000) in the table, with AdaBoost also strong (0.829), and PCA at 0.300 — some methods detect most anomalies while others miss many.

AdaBoost leads with F1 = 0.874, showing the best balance between precision and recall on ECG; PCA and OC-SVM produce moderate F1s.

Method	Precision	Recall	F1-score
AdaBoost	0.923	0.829	0.874
Autoencoder	0.585	0.305	0.401
DBSCAN	0.096	1.000	0.175
Elliptic Env.	0.489	0.255	0.335
GMM	0.250	0.130	0.171
Isolation Forest	0.494	0.257	0.338
K-Means	0.088	0.046	0.061
KNN	0.422	0.219	0.289
LOF	0.214	0.111	0.146
OC-SVM	0.546	0.285	0.374
PCA	0.576	0.300	0.395

Table 6.1: Performance on ECG (MIT-BIH) dataset

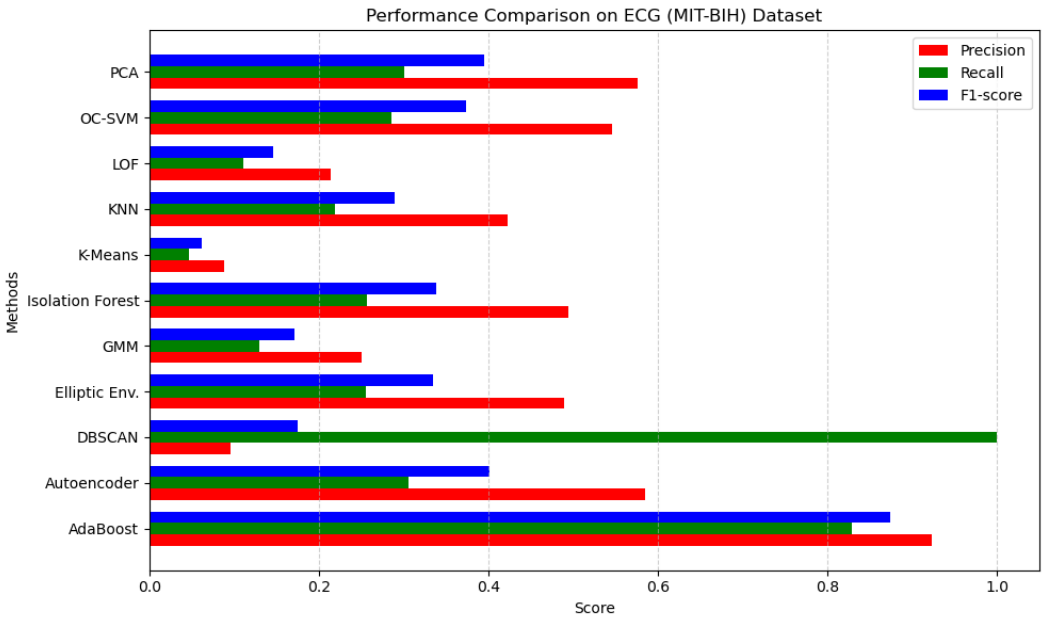


Figure 6.1: Precision, recall, and F1-score performance on MIT-BIT dataset

6.2 TEMPERATURE (NAB)

DBSCAN has the highest precision (0.917), indicating few false positives when it flags anomalies, but other methods (AdaBoost 0.615, PCA 0.378) show moderate precision. Recall values are low overall; the top recall is Autoencoder at 0.244, but most methods are under 0.25, indicating many anomalies are missed.

F1 is low across methods; Autoencoder is highest among listed (0.324), but still modest overall detection quality for temperature anomalies is weak.

Method	Precision	Recall	F1-score
AdaBoost	0.615	0.140	0.229
Autoencoder	0.486	0.244	0.324
DBSCAN	0.917	0.061	0.114
Elliptic Env.	0.343	0.172	0.229
GMM	0.437	0.219	0.292
Isolation Forest	0.456	0.229	0.305
K-Means	0.445	0.223	0.297
KNN	0.404	0.202	0.270
LOF	0.179	0.090	0.119
OC-SVM	0.263	0.202	0.229
PCA	0.378	0.090	0.145

Table 6.2: Performance on Temperature dataset

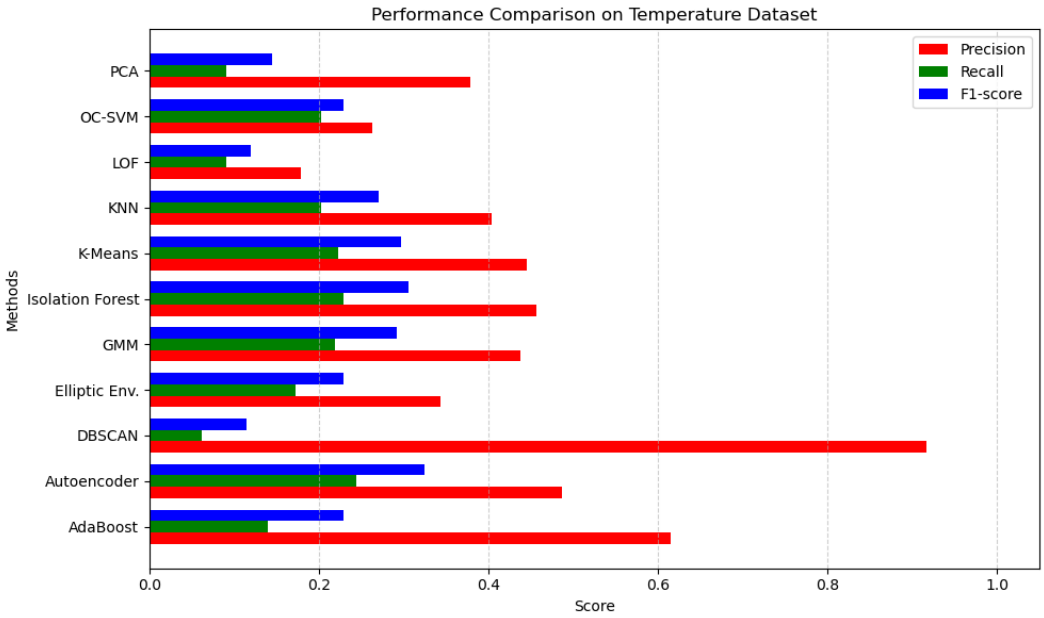


Figure 6.2: Precision, recall, and F1-score performance on Temperature (NAB)

6.3 SMART-HOME FALL DETECTION DATASET

Precision is uniformly low; the top value is AdaBoost at 0.291 and Elliptic Envelope at 0.239. This indicates many flagged events are false positives.

Best recall is Elliptic Envelope (0.244) and PCA (0.152); recall is low across all methods, indicating that a large proportion of fall events are not detected.

F1 scores are low; Elliptic Envelope gives the best F1 (0.241), but performance remains poor overall — fall detection is a challenging task across models.

Method	Precision	Recall	F1-score
AdaBoost	0.291	0.004	0.009
Autoencoder	0.049	0.050	0.049
DBSCAN	0.072	0.110	0.087
Elliptic Env.	0.239	0.244	0.241
GMM	0.122	0.125	0.123
Isolation Forest	0.144	0.147	0.146
K-Means	0.034	0.032	0.033
KNN	0.079	0.081	0.080
LOF	0.080	0.082	0.081
OC-SVM	0.077	0.079	0.078
PCA	0.149	0.152	0.151

Table 6.3: Performance on Fall Detection dataset

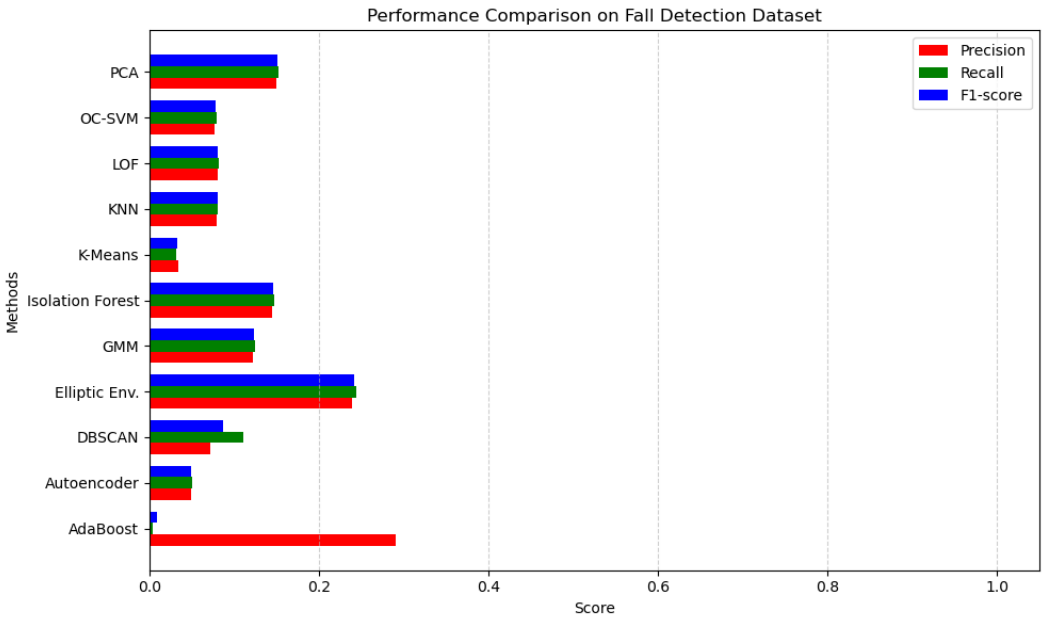


Figure 6.3: Precision, recall, and F1-score performance on fall detection dataset

6.4 FINANCIAL FRAUD TRANSACTION DATASET

Precision is very high for AdaBoost (0.976) and strong for DBSCAN (0.808) and PCA (0.659), meaning fraud detections are, in general, precise under those methods.

Several methods show extremely high recall — DBSCAN and GMM at 1.000, and AdaBoost at 0.985 — indicating fraud cases are being captured comprehensively by these methods.

AdaBoost achieves an outstanding F1 (0.980), with DBSCAN and GMM also high (0.894 and 0.805, respectively) — and most methods do very well on financial fraud detection.

Method	Precision	Recall	F1-score
AdaBoost	0.976	0.985	0.980
Autoencoder	0.529	0.033	0.062
DBSCAN	0.808	1.000	0.894
Elliptic Env.	0.117	0.007	0.014
GMM	0.529	0.032	0.062
Isolation Forest	0.339	0.021	0.040
K-Means	0.000	0.000	0.000
KNN	0.620	0.038	0.072
LOF	0.716	0.044	0.083
OC-SVM	0.510	0.033	0.063
PCA	0.659	0.041	0.077

Table 6.4: Performance on Financial Fraud dataset

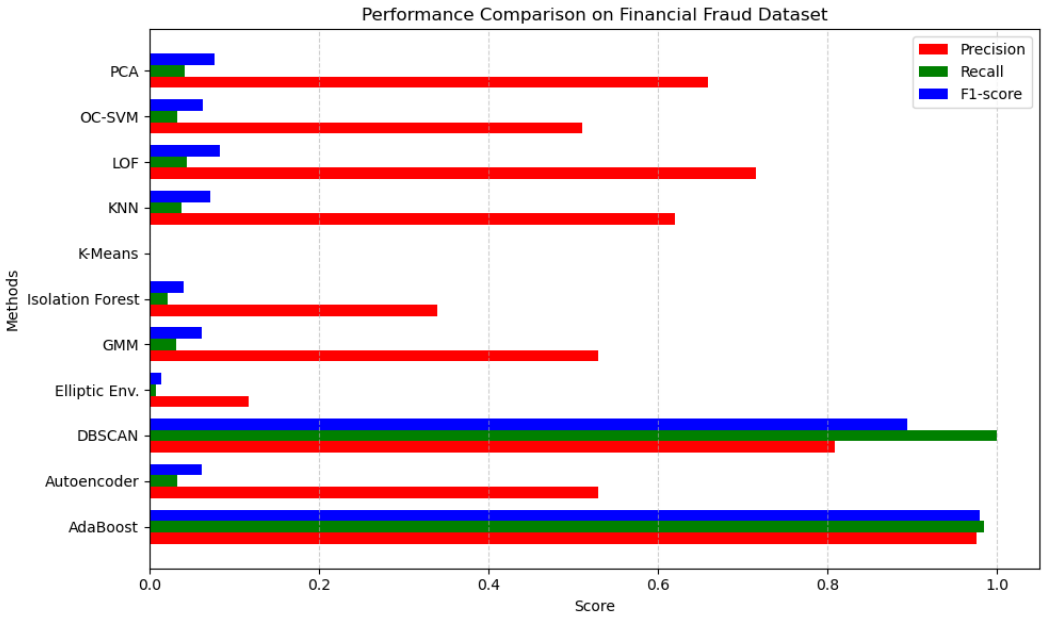


Figure 6.4: Precision, recall, and F1-score performance on Financial Fraud dataset

6.5 TEXTILE FABRIC DEFECT DATASET (AITEX)

AdaBoost again shows perfect precision (1.000); DBSCAN (0.623), Elliptic Envelope (0.616), and PCA (0.588) show moderate precision.

Recall is very high for DBSCAN (1.000), Elliptic Envelope (0.972), and GMM (0.958) — defects are generally captured well.

DBSCAN (0.768), Elliptic Envelope (0.754), and GMM (0.747) deliver the best F1s, indicating reliable defect detection for those density/covariance approaches; Autoencoder F1 (0.324) is relatively low.

Method	Precision	Recall	F1-score
AdaBoost	1.000	0.099	0.179
Autoencoder	0.338	0.309	0.324
DBSCAN	0.623	1.000	0.768
Elliptic Env.	0.616	0.972	0.754
GMM	0.613	0.958	0.747
Isolation Forest	0.320	0.239	0.274
K-Means	0.636	0.493	0.556
KNN	0.500	0.042	0.078
LOF	0.764	0.183	0.295
OC-SVM	0.538	0.690	0.605
PCA	0.588	0.845	0.694

Table 6.5: Performance on AITEX Fabric dataset

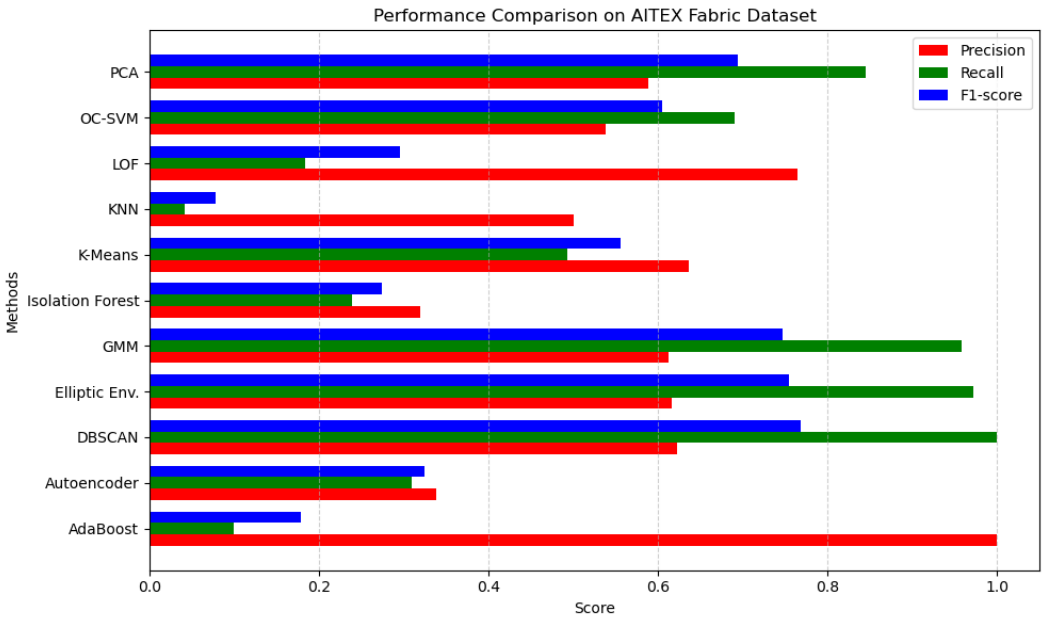


Figure 6.5: Precision, recall, and F1-score performance on AITEX Fabric dataset

6.6 BRAIN MRI ANOMALY DATASET

AdaBoost reports perfect precision (1.000); PCA (0.707) and GMM (0.691) also show good precision. While precision on test data is 1, its performance on new data may be lower. Very high recall for GMM (0.966), DBSCAN (1.000), and Elliptic Envelope (0.896); PCA is also high (0.828) — most tumor cases are detected by several methods. PCA attains the top F1 (0.829) with GMM (0.805) and DBSCAN (0.806) close behind — image anomalies (tumors) are well captured by these methods.

Method	Precision	Recall	F1-score
AdaBoost	1.000	0.121	0.215
Autoencoder	0.608	0.241	0.346
DBSCAN	0.674	1.000	0.806
Elliptic Env.	0.694	0.896	0.782
GMM	0.691	0.966	0.805
Isolation Forest	0.615	0.138	0.225
K-Means	0.677	0.362	0.472
KNN	0.500	0.034	0.064
LOF	0.500	0.068	0.121
OC-SVM	0.543	0.431	0.480
PCA	0.707	0.828	0.829

Table 6.6: Performance on Brain MRI dataset

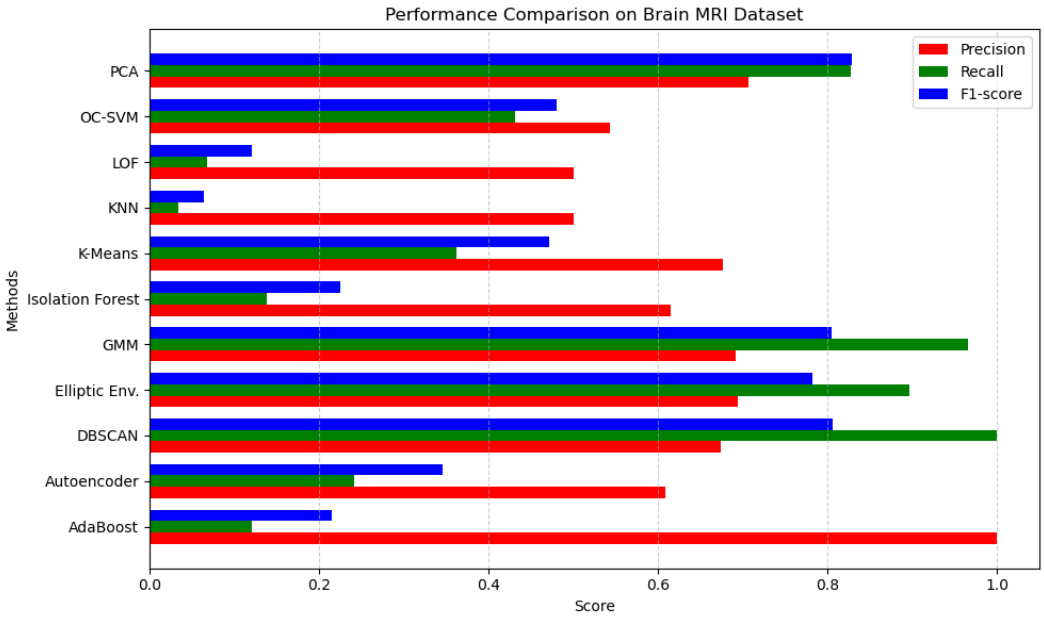


Figure 6.6: Precision, recall, and F1-score performance on Brain MRI dataset

6.7 OFFENSIVE LANGUAGE IDENTIFICATION DATASET

AdaBoost has the highest precision (0.658), followed by KNN and LOF shows that false-positive control is clearly best under the supervised pipeline.

AdaBoost achieves the highest recall (0.450), followed by LOF (0.350), Elliptic Envelope (0.350), and KNN (0.338). Even the best recall is modest, showing that OLID remains difficult for all methods.

AdaBoost has the highest F1-score (0.535), followed by LOF (0.325), KNN (0.324), and Elliptic Envelope (0.302). This means AdaBoost remains clearly best once both precision and recall are balanced.

The results mentioned in this section are when the sing classic natural language processing (NLP) feature extractions like Term Frequency-Inverse Document Frequency (TF-IDF) and etc is used as mentioned in section 5.2 as feature extraction setting. The other results with different feature setting are mentioned in appendix A.1 section.

Method	Precision	Recall	F1-score
KNN	0.311	0.338	0.324
OC-SVM	0.277	0.304	0.290
Autoencoder	0.271	0.292	0.281
Iso. Forest	0.261	0.283	0.272
AdaBoost	0.658	0.450	0.535
LOF	0.303	0.350	0.325
PCA	0.271	0.292	0.281
Elliptic Env.	0.266	0.350	0.302
DBSCAN	0.287	0.308	0.297
GMM	0.269	0.287	0.278
K-Means	0.267	0.287	0.277

Table 6.7: Performance on OLID dataset

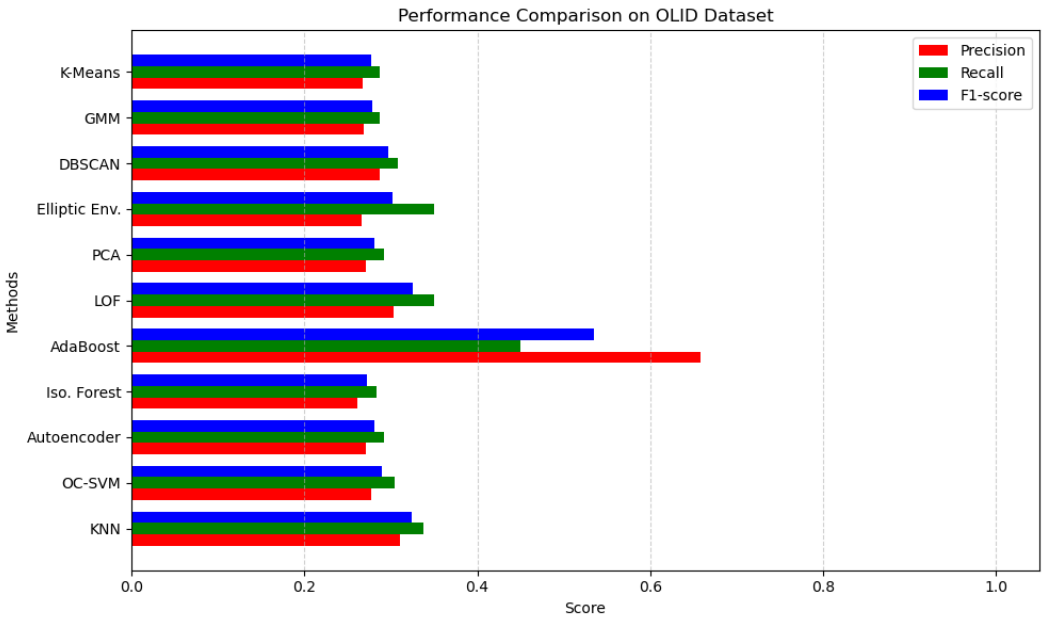


Figure 6.7: Precision, recall, and F1-score performance on OLID dataset

6.8 IMDB LARGE MOVIE REVIEW DATASET

AdaBoost the highest precision (0.793), followed by Elliptic Envelope, and KNN showing moderate precision.

DBSCAN reaches the highest recall (1.000), followed by GMM (0.963) and AdaBoost (0.793). This indicates that some methods detect almost all anomalies on IMDB, although not always with the best precision.

AdaBoost again leads with the highest F1-score (0.793), followed by DBSCAN (0.667), GMM (0.664), and One-Class SVM (0.193). The gap shows that AdaBoost is much more balanced than the remaining methods on IMDB.

Method	Precision	Recall	F1-score
KNN	0.519	0.052	0.094
OC-SVM	0.480	0.120	0.193
Autoencoder	0.477	0.048	0.087
Iso. Forest	0.510	0.046	0.084
AdaBoost	0.793	0.793	0.793
LOF	0.448	0.038	0.070
PCA	0.513	0.051	0.093
Elliptic Env.	0.523	0.048	0.087
DBSCAN	0.500	1.000	0.667
GMM	0.507	0.963	0.664
K-Means	0.475	0.048	0.086

Table 6.8: Performance on IMDB dataset

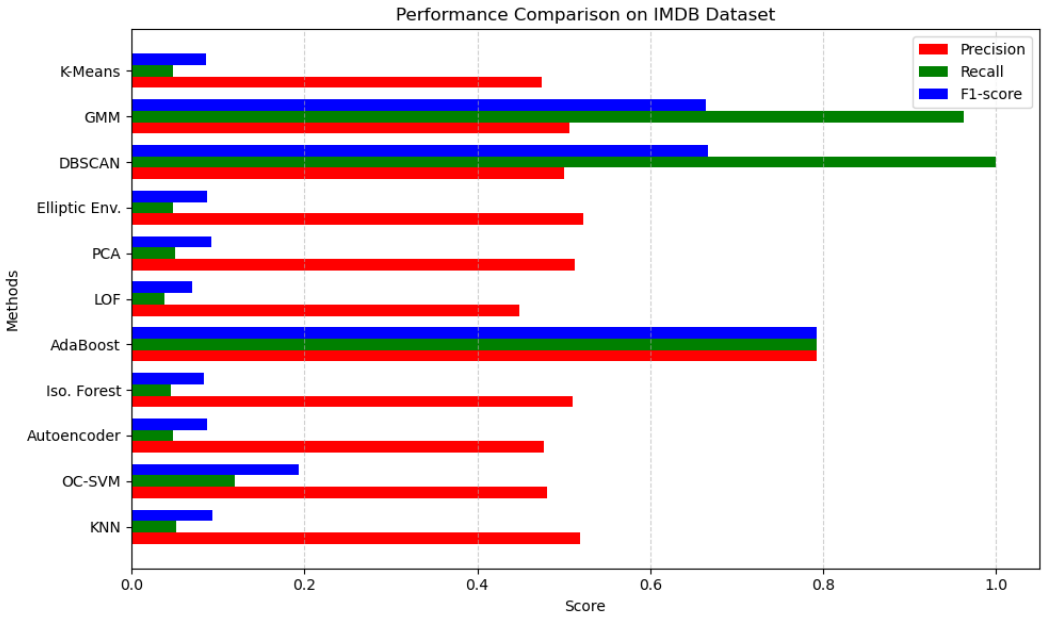


Figure 6.8: Precision, recall, and F1-score performance on IMDB dataset

6.9 UCSD ANOMALY DETECTION DATASET

KNN, Autoencoder, and PCA all reach perfect precision (1.000). However, these perfect-precision cases do not correspond to the strongest overall F1-score, indicating very conservative anomaly flags.

Elliptic Envelope and DBSCAN both reach recall 1.000, while LOF follows closely at 0.983 and One-Class SVM reaches 0.791.

Elliptic Envelope, DBSCAN, and LOF all reach the same top F1-score (0.901), while One-Class SVM follows at 0.846. This indicates that several unsupervised detectors are equally competitive on UCSD.

The results mentioned in this section are when the ResNet-based frame-level is used as feature extraction setting. The other results with different feature setting are mentioned in appendix A.3 section.

Method	Precision	Recall	F1-score
KNN	1.000	0.061	0.115
OC-SVM	0.908	0.791	0.846
Autoencoder	1.000	0.061	0.115
Iso. Forest	0.927	0.337	0.494
AdaBoost	0.000	0.000	0.000
LOF	0.831	0.983	0.901
PCA	1.000	0.061	0.115
Elliptic Env.	0.820	1.000	0.901
DBSCAN	0.820	1.000	0.901
GMM	0.954	0.349	0.511
K-Means	1.000	0.061	0.115

Table 6.9: Performance on UCSD dataset

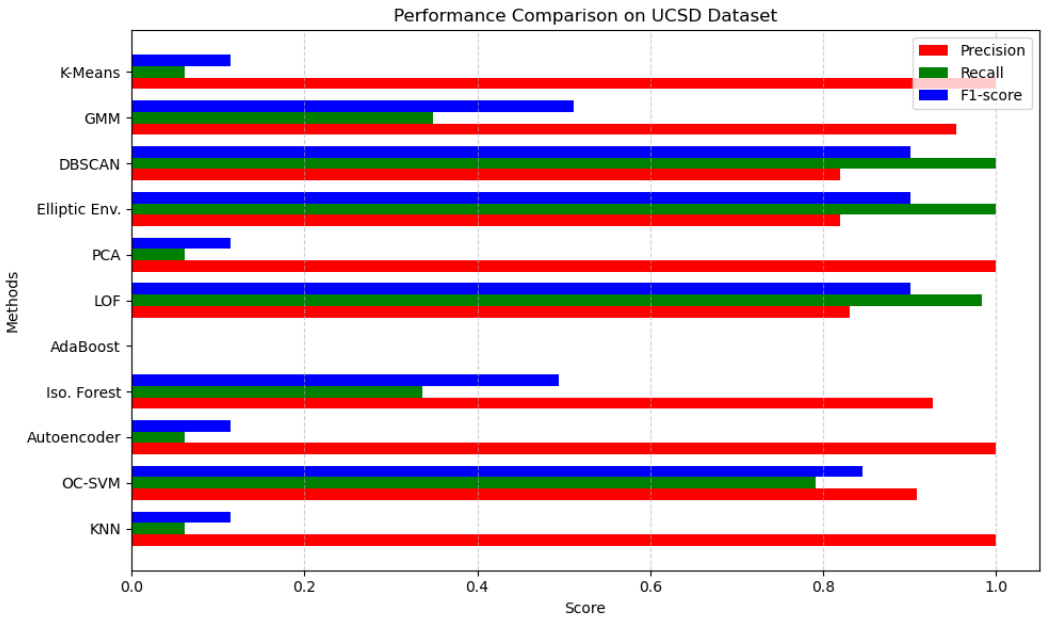


Figure 6.9: Precision, recall, and F1-score performance on UCSD dataset

6.10 UCF CRIME DATASET

Isolation Forest has the highest precision (0.935), closely followed by One-Class SVM (0.934), DBSCAN (0.926), and Autoencoder (0.916). AdaBoost is slightly lower at 0.826, but it later proves much stronger once recall is considered. AdaBoost has the highest recall by a large margin (1.000), whereas the next-best values are much lower: Isolation Forest 0.153, One-Class SVM 0.148, and DBSCAN 0.133. This shows that AdaBoost performs well in detecting anomalies on UCF-Crime. AdaBoost has the highest F1-score (0.905), far ahead of Isolation Forest (0.262), One-Class SVM (0.256), and DBSCAN (0.232). The dominance of AdaBoost is much clearer here than in the precision table alone.

The results mentioned in this section are when the 3D-convolutional is used as feature extraction setting. The other results with different feature setting are mentioned in appendix A.3 section.

Method	Precision	Recall	F1-score
KNN	0.899	0.094	0.170
OC-SVM	0.934	0.148	0.256
Autoencoder	0.916	0.115	0.204
Iso. Forest	0.935	0.153	0.262
AdaBoost	0.826	1.000	0.905
LOF	0.911	0.107	0.192
PCA	0.901	0.096	0.173
Elliptic Env.	0.762	0.034	0.065
DBSCAN	0.926	0.133	0.232
GMM	0.884	0.080	0.147
K-Means	0.910	0.106	0.190

Table 6.10: Performance on UCF-Crime dataset

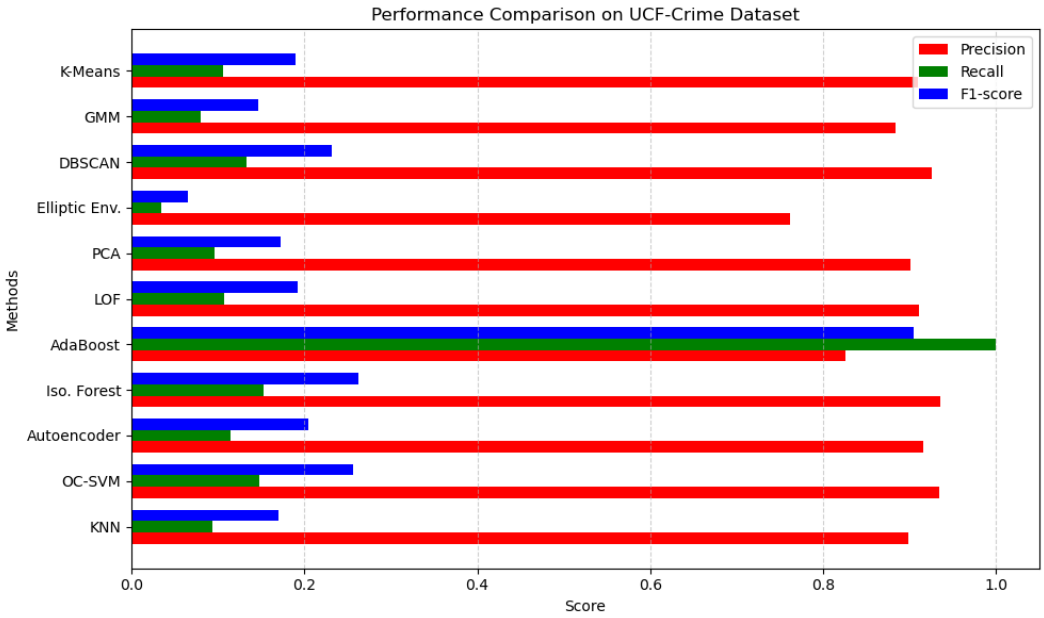


Figure 6.10: Precision, recall, and F1-score performance on UCF-Crime dataset

6.11 MIMII DATASET

Under the retained combined-only setting, GMM reaches the highest precision (0.918), followed by Autoencoder (0.804), PCA (0.709), and DBSCAN (0.708). LOF is lower in precision (0.532), but it is more balanced across metrics.

LOF attains the highest recall (0.568), followed by One-Class SVM (0.309), Elliptic Envelope (0.270), and Isolation Forest (0.253). The overall values remain low, so missed anomalies are still common in the combined-only MIMII setting.

LOF achieves the highest F1-score (0.549), followed by Elliptic Envelope (0.357), One-Class SVM (0.328), and GMM (0.292). This confirms that, under the combined-only selection rule, LOF is the strongest balanced method on MIMII.

The results mentioned in this section are when the log-Mel/MFCC statistics is used as feature extraction setting. The other results with different feature setting are mentioned in appendix A.2 section.

Method	Precision	Recall	F1-score
KNN	0.651	0.123	0.207
OC-SVM	0.348	0.309	0.328
Autoencoder	0.804	0.152	0.256
Iso. Forest	0.293	0.253	0.272
AdaBoost	0.162	0.004	0.007
LOF	0.532	0.568	0.549
PCA	0.709	0.134	0.226
Elliptic Env.	0.527	0.270	0.357
DBSCAN	0.708	0.134	0.225
GMM	0.918	0.174	0.292
K-Means	0.426	0.081	0.136

Table 6.11: Performance on MIMII dataset

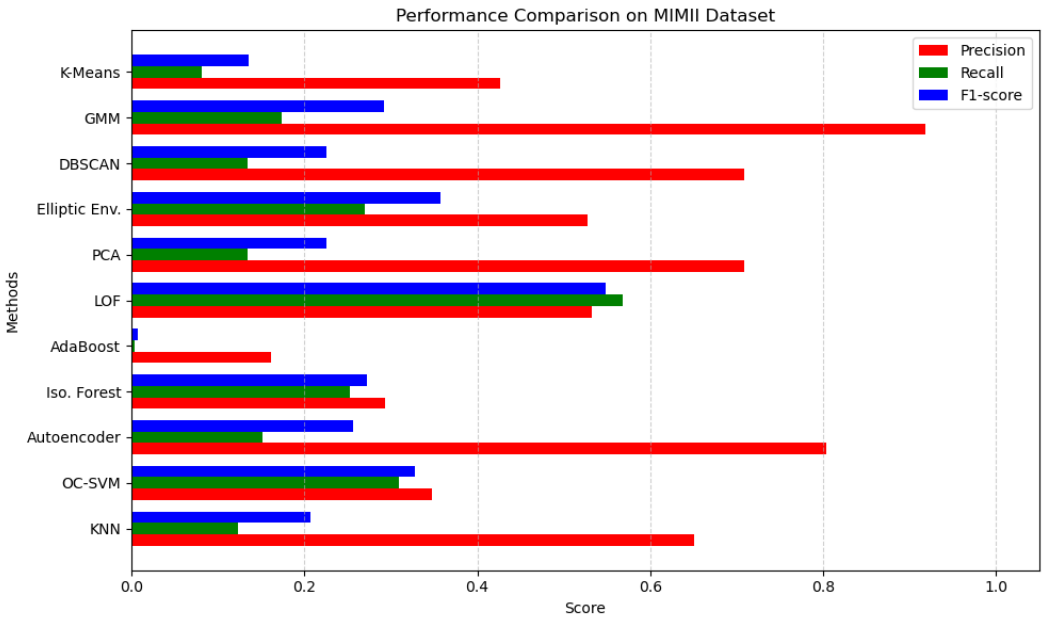


Figure 6.11: Precision, recall, and F1-score performance on MIMII dataset

6.12 ASVSPOOF 2019 DATASET

Precision is very high across several methods. GMM leads with 0.984, followed by One-Class SVM (0.981), K-Means (0.976), PCA (0.968), and AdaBoost (0.967). This means false alarms can be controlled well on this benchmark even before optimizing the precision–recall trade-off. DBSCAN reaches the highest recall (1.000), with LOF close behind at 0.976 and KNN at 0.957. This indicates that the normal and anomalous data points in the dataset can be clearly and easily distinguished from one another.
DBSCAN attains the highest F1-score (0.946), followed closely by LOF (0.938), KNN (0.930), and Isolation Forest (0.901). ASVspoof is therefore one of the strongest overall audio benchmarks in the repository.

The results mentioned in this section are when the log-Mel/MFCC statistics is used as feature extraction setting. The other results with different feature setting are mentioned in appendix A.2 section.

Method	Precision	Recall	F1-score
KNN	0.905	0.957	0.930
OC-SVM	0.981	0.612	0.753
Autoencoder	0.966	0.553	0.704
Iso. Forest	0.934	0.869	0.901
AdaBoost	0.967	0.006	0.012
LOF	0.903	0.976	0.938
PCA	0.968	0.547	0.699
Elliptic Env.	0.924	0.829	0.874
DBSCAN	0.897	1.000	0.946
GMM	0.984	0.342	0.507
K-Means	0.976	0.371	0.538

Table 6.12: Performance on ASVspoof dataset

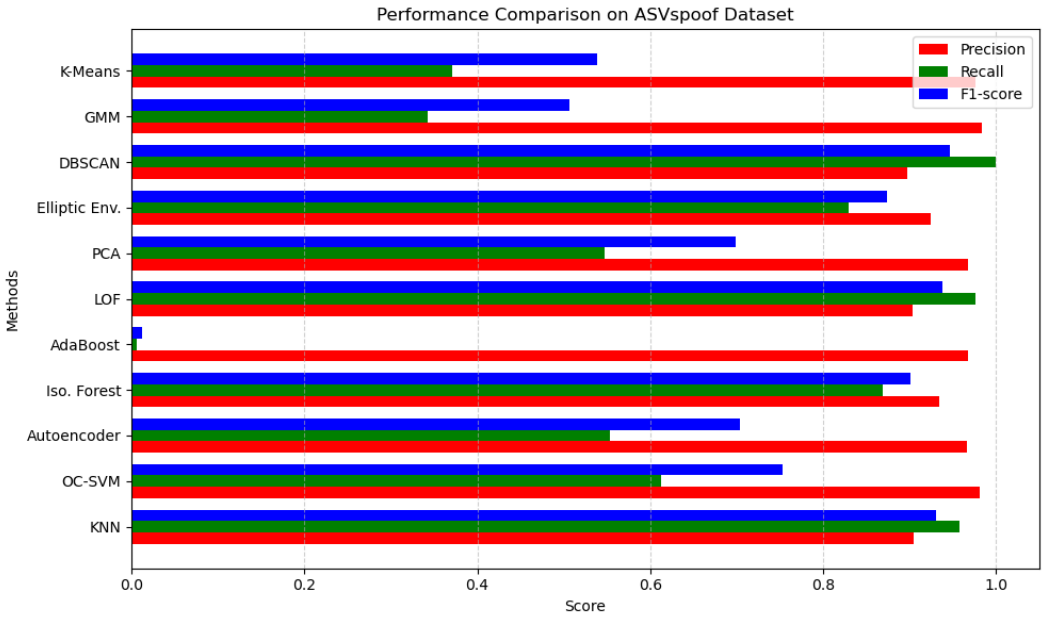


Figure 6.12: Precision, recall, and F1-score performance on ASVspoof dataset

6.13 RAID DATASET

Autoencoder has the highest precision (0.987), followed by Elliptic Envelope (0.983), One-Class SVM (0.980), and AdaBoost (0.977). Precision is therefore uniformly strong on RAID, even though the best F1-score later belongs to AdaBoost.

AdaBoost has the highest recall (0.999), followed by Elliptic Envelope (0.841), Autoencoder (0.498), and LOF (0.496). AdaBoost performes well in detecting anomaly on RAID dataset.

AdaBoost reaches the highest F1-score (0.988), followed by Elliptic Envelope (0.906), Autoencoder (0.662), and One-Class SVM (0.657). This shows that AdaBoost provides by far the best overall balance on RAID.

The results mentioned in this section are when the classic natural language processing (NLP) feature extractions like Term Frequency-Inverse Document Frequency (TF-IDF) and etc is used as mentioned in section 5.2 as feature extraction setting. The other results with different feature setting are mentioned in appendix A.1 section.

Method	Precision	Recall	F1-score
KNN	0.970	0.489	0.650
OC-SVM	0.980	0.494	0.657
Autoencoder	0.987	0.498	0.662
Iso. Forest	0.976	0.494	0.656
AdaBoost	0.977	0.999	0.988
LOF	0.965	0.496	0.655
PCA	0.993	0.501	0.666
Elliptic Env.	0.983	0.841	0.906
DBSCAN	0.972	0.490	0.652
GMM	0.976	0.492	0.654
K-Means	0.975	0.492	0.654

Table 6.13: Performance on RAID dataset

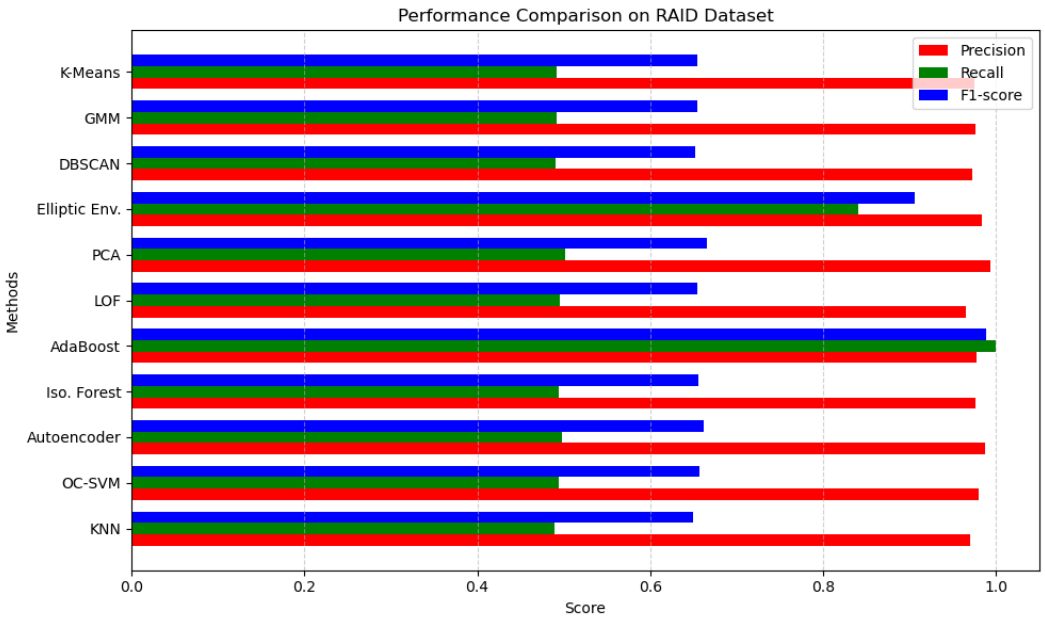


Figure 6.13: Precision, recall, and F1-score performance on RAID dataset

6.14 AI VS HUMAN TEXT

KNN achieves the highest precision (0.990), followed by LOF (0.906), AdaBoost (0.825), and GMM (0.739). This indicates that the LLM-embedding representation is highly effective for minimizing false alarms

KNN achieves the highest recall (0.993), followed by PCA (0.949), Autoencoder (0.945), and One-Class SVM (0.899). This confirms that the AI-vs-human benchmark is highly separable in the retained representation.

KNN has the highest F1-score (0.991), while AdaBoost and LOF are tied at 0.789 and GMM follows at 0.768. KNN is be a better method to choose to detect anomaly.

This following are the results which were obtained after using feature extraction methods based on transformer-feature setting. The rest of the results which were obtained by different feature extraction setting are mention in the appendix section A.1.

Method	Precision	Recall	F1-score
KNN	0.946	0.947	0.946
OC-SVM	0.538	0.901	0.537
Autoencoder	0.536	0.939	0.683
Iso. Forest	0.509	0.693	0.587
AdaBoost	0.861	0.837	0.849
LOF	0.804	0.691	0.743
PCA	0.522	0.945	0.672
Elliptic Env.	0.531	0.805	0.640
DBSCAN	0.818	0.011	0.021
GMM	0.854	0.807	0.829
K-Means	0.223	0.023	0.226

Table 6.14: Performance on AI vs Human dataset

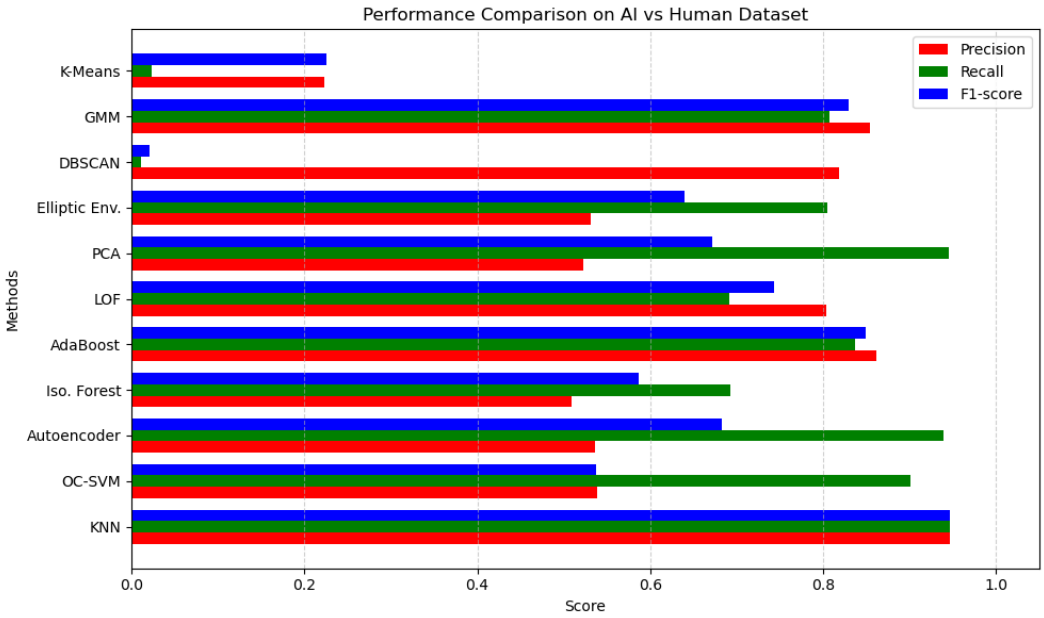


Figure 6.14: Precision, recall, and F1-score performance on AI-vs-human dataset

6.15 TRANSFER PERFORMANCE FOR MACHINE GENERATED TEXT

This work has conducted few experiments on transfer performance for machine generated text. Transfer performance refer to measuring of how effectively a model, pre-trained on a source data, applies its learned knowledge to a new, related target data.

RAID → AI vs Human Dataset:

This are the results of the transfer performance in which the models where trained on the RADI dataset and evaluated on the AI vs Human Dataset. This will help in understanding how does the learning of models on the RAID dataset can be helpful in evaluating this models on similar task or dataset.

Autoencoder has the highest precision (0.981), followed by Elliptic Envelope (0.979), Isolation Forest (0.978), and One-Class SVM (0.978). This direction produces very high precision, although the best balanced model is AdaBoost. AdaBoost has the highest recall (0.997), followed by Elliptic Envelope (0.829), Autoencoder (0.495), and One-Class

SVM (0.493). This means the RAID to AI vs Human Dataset transfer direction still allows very high anomaly coverage for the strongest method. AdaBoost reaches the highest F1-score (0.987), followed by Elliptic Envelope (0.898), Autoencoder (0.658), and One-Class SVM (0.656). So the RAID to AI vs Human Dataset direction remains strong, but not as strong as the reverse direction.

This results are obtained by using classic natural language processing (NLP) feature extractions like Term Frequency-Inverse Document Frequency (TF-IDF) and etc. as mentioned in section 5.2, other feature extraction settings results are mentioned in appendix section A.1.

Method	Precision	Recall	F1-score
KNN	0.975	0.492	0.654
OC-SVM	0.978	0.493	0.656
Autoencoder	0.981	0.495	0.658
Iso. Forest	0.978	0.485	0.648
AdaBoost	0.978	0.997	0.987
LOF	0.963	0.486	0.646
PCA	0.991	0.500	0.664
Elliptic Env.	0.979	0.829	0.898
DBSCAN	0.972	0.490	0.652
GMM	0.972	0.490	0.652
K-Means	0.974	0.491	0.653

Table 6.15: Performance on RAID → AI vs Human Dataset

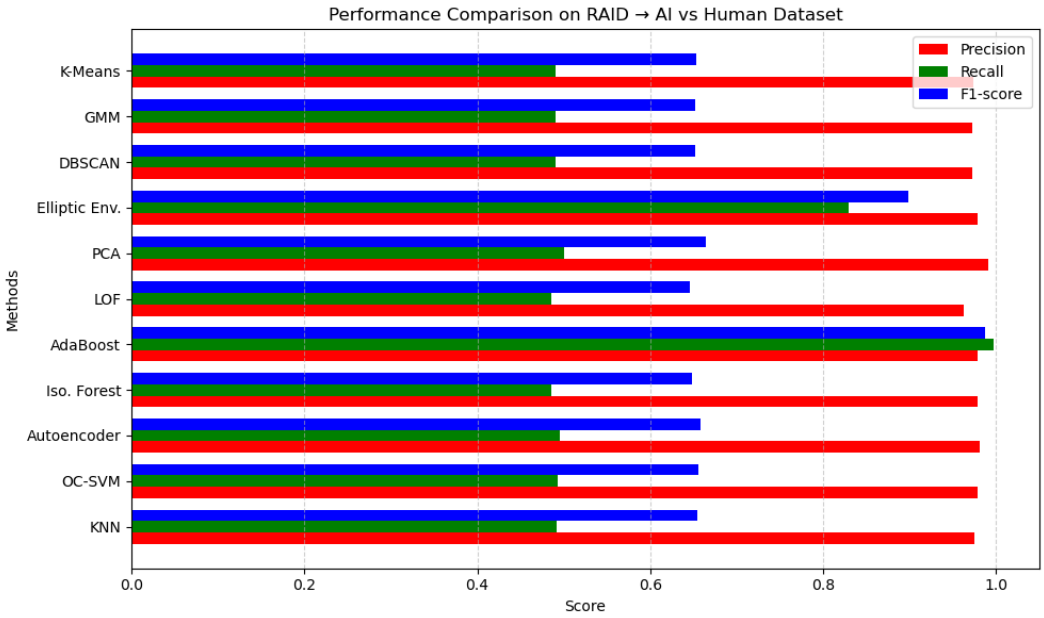


Figure 6.15: Precision, recall, and F1-score performance on RAID → AI vs Human Dataset

AI vs Human Dataset → RAID:

This are the results of the transfer performance in which the models where trained on the AI vs Human Dataset dataset and evaluated on the RAID Dataset. This will help in understanding how does the learning of models on the AI vs Human Dataset dataset can be helpful in evaluating this models on similar task or dataset.

KNN again has the highest precision (0.990), followed by LOF (0.906), AdaBoost (0.825), and GMM (0.739). This direction remains highly separable in the retained representative results.

KNN again has the highest recall (0.993), followed by PCA (0.949), Autoencoder (0.945), and One-Class SVM (0.899). The Kaggle-to-RAID transfer setting therefore preserves very strong anomaly coverage.

KNN again reaches the highest F1-score (0.991), followed by AdaBoost (0.789), LOF (0.789), and GMM (0.768). The AI vs Human Dataset to RAID transfer experiment

therefore mirrors the direct AI-vs-human ranking very closely.

This are the results when transformer is used for feature extraction to get embeddings. The other feature extraction settings results are mentioned in the appendix section A.1.

Method	Precision	Recall	F1-score
KNN	0.946	0.947	0.946
OC-SVM	0.528	0.901	0.665
Autoencoder	0.536	0.939	0.683
Iso. Forest	0.509	0.693	0.587
AdaBoost	0.861	0.837	0.849
LOF	0.804	0.691	0.743
PCA	0.522	0.945	0.672
Elliptic Env.	0.531	0.805	0.640
DBSCAN	0.818	0.011	0.021
GMM	0.854	0.807	0.829
K-Means	0.223	0.023	0.226

Table 6.16: Performance on AI vs Human → RAID

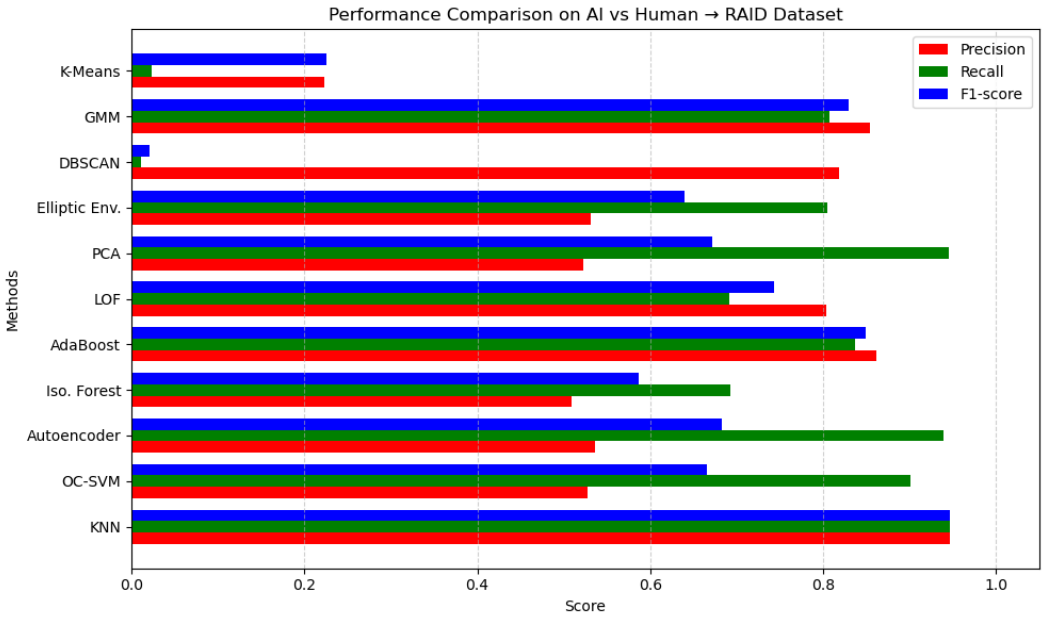


Figure 6.16: Precision, recall, and F1-score performance on AI vs Human → RAID

6.16 COMBINED MULTIPLE DATASETS IN MACHINE GENERATED TEXT

In this work, multiple datasets where combined to evaluate performance of different models. The dataset includes the AI-vs-Human dataset, the RAID dataset, the M-DAIGT dataset, CT2 dataset, and AIGTxt dataset.

AdaBoost reports the highest precision (0.991), while Autoencoder (0.975), PCA (0.974), and DBSCAN (0.973) also remain very strong. This is one of the clearest high-precision settings in the repository.

AdaBoost leads with recall 0.996, followed by GMM (0.746), One-Class SVM (0.684), and Elliptic Envelope (0.555). This is another example where AdaBoost captures almost all anomalies.

AdaBoost achieves the highest F1-score (0.993), followed by GMM (0.752), Elliptic Envelope (0.703), and One-Class SVM (0.689). This makes the combined benchmark the strongest overall AdaBoost result in the repository.

The results mentioned in this section are from large language models (LLM) embeddings

feature extraction setting. The other results from different feature extraction setting are mention in appendix section A.1.

Method	Precision	Recall	F1-score
KNN	0.960	0.864	0.910
OC-SVM	0.505	0.900	0.647
Autoencoder	0.461	0.946	0.620
Iso. Forest	0.645	0.701	0.672
AdaBoost	0.795	0.783	0.789
LOF	0.420	0.696	0.524
PCA	0.441	0.950	0.603
Elliptic Env.	0.515	0.793	0.625
DBSCAN	0.577	0.336	0.425
GMM	0.755	0.766	0.761
K-Means	0.261	0.398	0.315

Table 6.17: Performance across combined multiple datasets

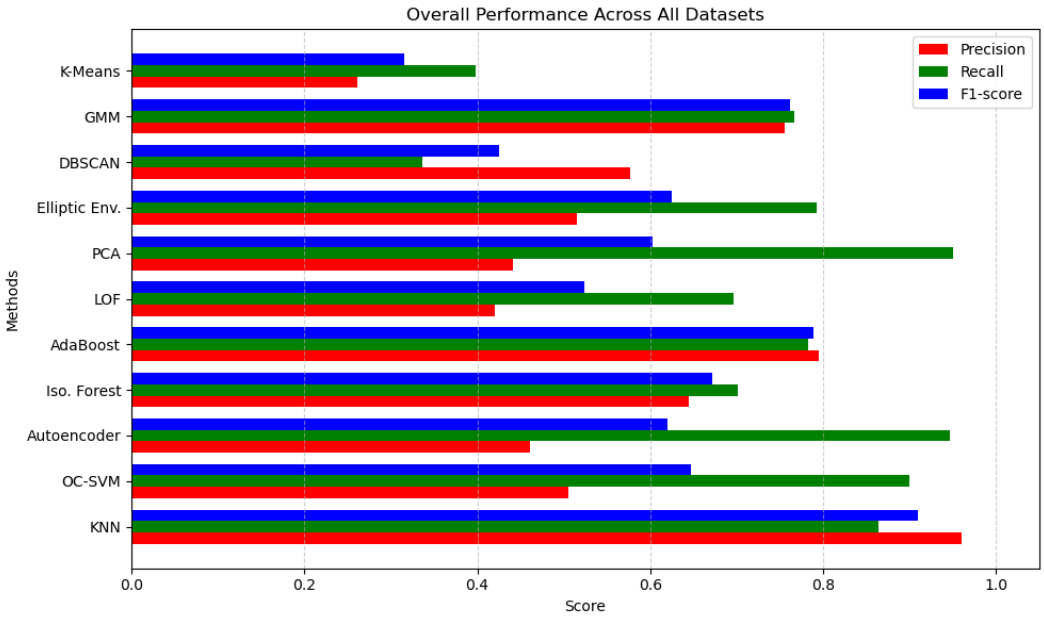


Figure 6.17: Precision, recall, and F1-score performance across combined multiple datasets

6.17 WILDDEEPFAKE DATASET

AdaBoost has the highest precision (0.685), followed by K-Means (0.645), GMM (0.620), and PCA (0.593). Precision is only moderate here, which indicates that false alarms remain a challenge on wildDeepFake.

DBSCAN achieves the highest recall (0.999), followed by LOF (0.709), Autoencoder (0.584), and KNN (0.490). The gap shows that DBSCAN is particularly suited to find anomalies on this dataset.

DBSCAN has the highest F1-score (0.713), followed by LOF (0.597), Autoencoder (0.552), and AdaBoost (0.548). Even the best value is only moderate, so this remains a comparatively difficult benchmark.

The results mentioned in this section are when the 3D-convolutional is used as feature extraction setting. The other results with different feature setting are mentioned in appendix A.3 section.

Method	Precision	Recall	F1-score
KNN	0.544	0.519	0.532
OC-SVM	0.509	0.134	0.212
Autoencoder	0.557	0.132	0.213
Iso. Forest	0.535	0.471	0.501
AdaBoost	0.615	0.526	0.568
LOF	0.523	0.607	0.562
PCA	0.439	0.061	0.107
Elliptic Env.	0.520	0.468	0.493
DBSCAN	0.590	0.790	0.676
GMM	0.516	0.080	0.139
K-Means	0.708	0.349	0.467

Table 6.18: Performance on wildDeepFake dataset

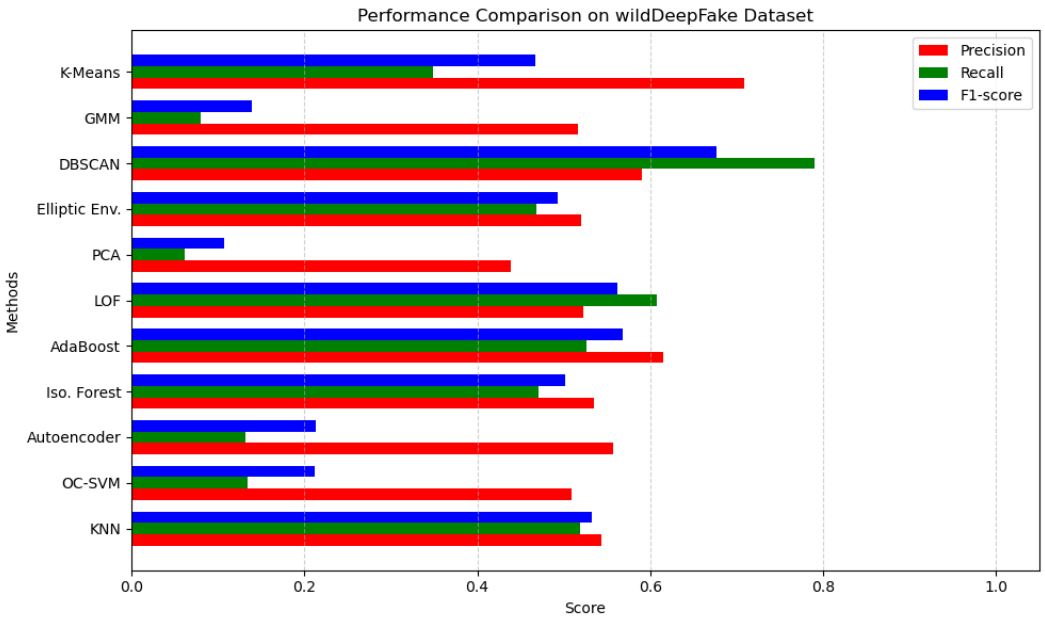


Figure 6.18: Precision, recall, and F1-score performance on wildDeepFake datasets

6.18 CELEB-DF (V2)

PCA has the highest precision (0.927), followed by GMM (0.917), LOF (0.910), and One-Class SVM (0.910).

DBSCAN and K-Means both reach perfect recall (1.000), while AdaBoost is nearly identical at 0.999 and KNN remains very high at 0.995.

KNN leads with the highest F1-score (0.932), while AdaBoost, DBSCAN, and K-Means are all very close at 0.927. Celeb-DF is therefore both stronger and more competitive across methods than wildDeepFake.

The results mentioned in this section are when the vision transformer is used as feature extraction setting. The other results with different feature setting are mentioned in appendix A.3 section.

Method	Precision	Recall	F1-score
KNN	0.876	0.995	0.932
OC-SVM	0.910	0.215	0.348
Autoencoder	0.684	0.338	0.453
Iso. Forest	0.896	0.453	0.602
AdaBoost	0.865	0.999	0.927
LOF	0.910	0.518	0.661
PCA	0.927	0.109	0.194
Elliptic Env.	0.877	0.550	0.676
DBSCAN	0.863	1.000	0.927
GMM	0.917	0.062	0.117
K-Means	0.863	1.000	0.927

Table 6.19: Performance on Celeb-DF dataset

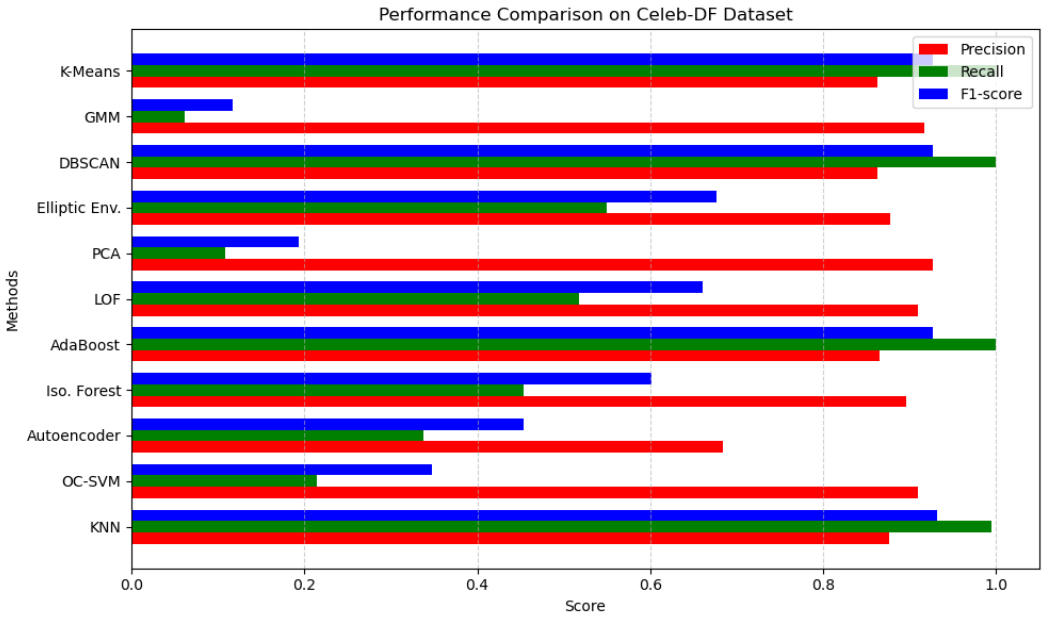


Figure 6.19: Precision, recall, and F1-score performance on Celeb-DF dataset

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

This thesis presents a broad comparative study of anomaly detection across time-series, tabular data, images, audio, text, surveillance video, machine-generated text, and deepfake video. The study does not identify a single universally best detector.

Across the full benchmark suite, AdaBoost emerges as the most consistent high-performing method when class boundaries are sufficiently learnable and some degree of supervision is available. This is evident in ECG, financial fraud, OLID, IMDB, UCF-Crime, RAID, and RAID-to-Kaggle transfer. In contrast, KNN performs exceptionally well when embeddings preserve semantic structure effectively, as seen in AI-generated text detection and Celeb-DF. Density- and covariance-oriented methods such as DBSCAN, GMM, PCA, and Elliptic Envelope perform best when normal behavior lies on compact manifolds or when anomalies appear as sparse deviations from dominant structure, which is especially visible in Brain MRI, AITEX, UCSD, ASVspoof, and wildDeepFake. The broader lesson is that good anomaly detection depends on matching model bias to data geometry rather than expecting one model family to solve every case.

The results also show that feature extraction is not a preprocessing detail; it is one of the main determinants of success. In several branches of the thesis, the difference between two feature pipelines is larger than the difference between two detectors applied to the same features. This is particularly clear in machine-generated text, deepfake video, and sequential data, where stronger embeddings or domain-aligned representations substantially improve downstream anomaly detection. Accordingly, the thesis demonstrates that anomaly detection should be understood as a full pipeline

problem involving representation learning, detector choice, score calibration, threshold selection, and evaluation under realistic class imbalance.

Finally, the hardest datasets in the study are as informative as the strongest ones. The weak results on Temperature and Fall Detection indicate that current feature spaces and detector families are still insufficient when anomalies are subtle, temporally diffuse, or strongly overlapped with normal behavior. These low-performing cases identify the main frontier for future work: not simply refining detectors, but building better representations and better sequential modeling strategies for difficult anomaly regimes.

Key Insights Across Modalities

1. **The results separate clearly into easy and hard anomaly-detection regimes.** Financial Fraud, RAID, RAID-to-Kaggle transfer, and the pooled machine-generated-text benchmark all achieve near-perfect best F1-scores (0.980, 0.988, 0.987, and 0.993 respectively), showing that some anomaly classes are highly separable once the representation is informative. In contrast, NAB Temperature and Fall Detection remain difficult for every method, with best F1-scores of only 0.324 and 0.241. This gap indicates that anomaly visibility, feature quality, and class overlap often matter more than the detector name alone.

2. **No single detector is universally optimal across all modalities.** AdaBoost dominates several structured or label-rich settings, including ECG (0.874), Financial Fraud (0.980), OLID (0.535), IMDB (0.793), UCF-Crime (0.905), RAID (0.988), and RAID-to-Kaggle (0.987). However, KNN becomes the strongest method on high-quality embedding spaces such as AI vs Human text (0.991) and Celeb-DF (0.932), while DBSCAN, PCA, and Elliptic Envelope are more effective on compact geometric or density-structured datasets such as AITEX, Brain MRI, and UCSD.

3. **Feature representation is repeatedly shown to be as important as detector choice.** The thesis results show substantial performance shifts when the feature pipeline changes, especially in text, video, and audio. The machine-generated-text benchmarks are the clearest example: once the representation becomes discriminative, even relatively simple detectors achieve very strong results. Conversely, weaker or less task-aligned features limit performance even when more sophisticated detectors are used.

4. **Classical methods remain highly competitive when matched well to the data geometry.** The strongest image results come from PCA, GMM, DBSCAN, and

Elliptic Envelope rather than from a generic autoencoder. Similarly, LOF is the best balanced method on MIMII, and DBSCAN performs best on ASVspoof and wildDeepFake. These findings show that classical density, neighborhood, and subspace-based methods still provide strong baselines and should not be treated as outdated.

5. **Real-world variability remains a major challenge.** The difference between wildDeepFake and Celeb-DF is particularly informative. Celeb-DF reaches a best F1-score of 0.932, while wildDeepFake peaks at only 0.676. This suggests that uncontrolled capture conditions, identity diversity, compression artifacts, and domain shift make real-world synthetic-media detection substantially harder than detection on cleaner benchmark data.

6. **Transfer performance is promising but asymmetric.** Cross-dataset machine-generated-text detection remains strong in both directions, but the best-performing detector changes with the source-target pair. RAID-to-Kaggle is best handled by AdaBoost, whereas Kaggle-to-RAID mirrors the direct AI vs Human setting and favors KNN. This indicates that transferability depends not only on the dataset but also on how the chosen representation shapes the geometry of normal and anomalous samples.

7.2 FUTURE WORK

Expand practical deployment questions: A useful extension is to evaluate inference cost, latency, explainability, and human-in-the-loop review. This is especially important for fraud detection, healthcare imaging, surveillance video, and synthetic-media detection, where practitioners not only need strong scores but also trustworthy and interpretable decision support.

7.3 FINAL REMARKS

The overall evidence of this thesis leads to a clear practical message: anomaly detection is not a single-model competition, but a system design problem. The best-performing solution depends on how normality is represented, how anomalies depart from that representation, and how the final score is converted into actionable decisions. The thesis shows that classical methods remain highly valuable, deep methods are most useful when paired with strong task-aware representations, and balanced evaluation is essential

for meaningful comparison. The most important takeaway is therefore not that one detector wins, but that successful anomaly detection requires aligning representation learning, model choice, calibration, and evaluation with the structure of the target domain. That perspective provides a stronger foundation for both future research and practical deployment.

APPENDIX A

ADDITIONAL RESULTS

This appendix intentionally includes *only* results that are not already tabulated in Chapter 6. The main chapter already reports the primary precision/recall/F1 comparison for Temperature, ECG, Falling People, Financial Fraud, AITEX, Brain MRI, and IMDB under the selected feature pipeline for each dataset. This appendix includes results which were obtained from alternative feature-extraction or representation pipelines that were evaluated but not shown in Chapter 6.

This organization keeps the appendix complementary to the main chapter instead of duplicating it.

A.1 TEXT AND MACHINE-GENERATED TEXT

OLID

Chapter 6 reports the classical-feature comparison. The tables below preserve the two omitted OLID pipelines: the TF-IDF baseline and sentence-embedding pipeline.

Table A.1: OLID results using the TF-IDF feature extractor.

Method	Precision	Recall	F1-score
KNN	0.274	0.279	0.276
LOF	0.274	0.279	0.276
One-Class SVM	0.270	0.358	0.308
Isolation Forest	0.262	0.287	0.274
Elliptic Envelope	0.212	0.242	0.226
PCA	0.271	0.292	0.281
GMM	0.260	0.279	0.269
K-Means	0.260	0.279	0.269
DBSCAN	0.287	0.308	0.297
Autoencoder	0.264	0.283	0.273
AdaBoost	0.875	0.117	0.206

Table A.2: OLID results using the all-MiniLM-L6-v2 embedding.

Method	Precision	Recall	F1-score
KNN	0.140	0.025	0.042
LOF	0.186	0.033	0.057
One-Class SVM	0.070	0.013	0.021
Isolation Forest	0.116	0.021	0.035
Elliptic Envelope	0.233	0.042	0.071
PCA	0.093	0.017	0.028
GMM	0.116	0.021	0.035
K-Means	0.093	0.017	0.028
Autoencoder	0.023	0.004	0.007
VAE	0.279	0.050	0.085

RAID Text

Chapter 6 keeps the classical-feature comparison for RAID. To avoid duplicating those tables, the appendix adds only supervised baselines that fall outside the shared Chapter 6 detector set, along with the omitted transformer and LLM-based runs.

Table A.3: Additional supervised RAID baselines on the classical-feature pipeline that were not included in Chapter 6.

Method	Precision	Recall	F1-score
Gradient Boosting (GBM)	0.979	0.999	0.989
XGBoost	0.996	0.876	0.932
LightGBM	0.996	0.875	0.932
CatBoost	0.996	0.855	0.920

Table A.4: RAID results using the transformer-embedding pipeline.

Method	Precision	Recall	F1-score
KNN	0.760	0.862	0.808
LOF	0.630	0.682	0.655
One-Class SVM	0.557	0.898	0.687
Isolation Forest	0.584	0.694	0.635
Elliptic Envelope	0.531	0.799	0.638
PCA	0.564	0.944	0.706
GMM	0.614	0.673	0.642
K-Means	0.456	0.564	0.504
DBSCAN	0.998	0.459	0.629
Autoencoder	0.576	0.936	0.713
AdaBoost	0.688	0.708	0.698

Table A.5: RAID results using the LLM-embedding pipeline.

Method	Precision	Recall	F1-score
KNN	0.903	0.674	0.772
LOF	0.047	0.709	0.089
One-Class SVM	0.038	0.898	0.073
Isolation Forest	0.040	0.704	0.076
Elliptic Envelope	0.031	0.805	0.060
PCA	0.036	0.948	0.070
GMM	0.051	0.721	0.095
K-Means	0.024	0.639	0.047
DBSCAN	0.029	0.655	0.056
Autoencoder	0.045	0.933	0.085
AdaBoost	0.062	0.737	0.114

AI vs Human Text

The main chapter uses the transformer-feature setting. The appendix therefore retains the omitted classical-feature and LLM-embedding comparisons.

Table A.6: AI vs Human results using the classical NLP feature pipeline.

Method	Precision	Recall	F1-score
Gradient Boosting (GBM)	0.973	0.942	0.957
XGBoost	0.962	0.961	0.962
LightGBM	0.959	0.961	0.960
CatBoost	0.950	0.955	0.953
KNN	0.483	0.483	0.483
LOF	0.325	0.324	0.324
One-Class SVM	0.626	0.626	0.626
Isolation Forest	0.519	0.518	0.518
Elliptic Envelope	0.583	0.870	0.698
PCA	0.834	0.834	0.834
GMM	0.352	0.352	0.352
K-Means	0.548	0.548	0.548
DBSCAN	0.414	0.414	0.414
Autoencoder	0.840	0.840	0.840
AdaBoost	0.954	0.950	0.952

Table A.7: AI vs Human results using the LLM-embedding pipeline.

Method	Precision	Recall	F1-score
KNN	0.990	0.993	0.991
LOF	0.906	0.698	0.789
One-Class SVM	0.383	0.899	0.537
Isolation Forest	0.403	0.697	0.510
Elliptic Envelope	0.435	0.800	0.564
PCA	0.370	0.949	0.532
GMM	0.739	0.799	0.768
K-Means	0.258	0.141	0.182
DBSCAN	0.324	0.555	0.409
Autoencoder	0.383	0.945	0.545
AdaBoost	0.825	0.756	0.789

Transfer Text: RAID→Kaggle and Kaggle→RAID

Chapter 6 keeps the classical transfer results for RAID→Kaggle and the transformer transfer results for Kaggle→RAID. The appendix below records the remaining transfer experiments that were not shown in the main chapter.

Table A.8: Additional supervised baselines for the RAID→Kaggle classical transfer pipeline not included in Chapter 6.

Method	Precision	Recall	F1-score
Gradient Boosting (GBM)	0.977	0.999	0.988
XGBoost	0.994	0.890	0.939
LightGBM	0.994	0.893	0.941
CatBoost	0.994	0.857	0.920

Table A.9: RAID→Kaggle results using the transformer-embedding transfer pipeline.

Method	Precision	Recall	F1-score
KNN	0.760	0.862	0.808
LOF	0.630	0.682	0.655
One-Class SVM	0.557	0.898	0.687
Isolation Forest	0.584	0.694	0.635
Elliptic Envelope	0.531	0.799	0.638
PCA	0.564	0.944	0.706
GMM	0.614	0.673	0.642
K-Means	0.456	0.564	0.504
DBSCAN	0.998	0.459	0.629
Autoencoder	0.576	0.936	0.713
AdaBoost	0.688	0.708	0.698

Table A.10: RAID→Kaggle results using the LLM-embedding transfer pipeline.

Method	Precision	Recall	F1-score
KNN	0.617	0.000	0.000
LOF	0.842	0.016	0.031
One-Class SVM	0.929	0.000	0.001
Isolation Forest	0.725	0.007	0.014
Elliptic Envelope	0.638	0.494	0.557
PCA	0.839	0.020	0.039
GMM	0.000	0.000	0.000
K-Means	0.628	1.000	0.771
DBSCAN	0.645	0.761	0.699
Autoencoder	0.949	0.008	0.017
AdaBoost	0.250	0.000	0.000

Table A.11: Kaggle→RAID results using the classical transfer pipeline.

Method	Precision	Recall	F1-score
Gradient Boosting (GBM)	0.973	0.942	0.957
XGBoost	0.962	0.961	0.962
LightGBM	0.959	0.961	0.960
CatBoost	0.950	0.955	0.953
KNN	0.483	0.483	0.483
LOF	0.325	0.324	0.324
One-Class SVM	0.626	0.626	0.626
Isolation Forest	0.519	0.518	0.518
Elliptic Envelope	0.583	0.870	0.698
PCA	0.834	0.834	0.834
GMM	0.352	0.352	0.352
K-Means	0.548	0.548	0.548
DBSCAN	0.414	0.414	0.414
Autoencoder	0.840	0.840	0.840
AdaBoost	0.954	0.950	0.952

Table A.12: Kaggle→RAID results using the LLM-embedding transfer pipeline.

Method	Precision	Recall	F1-score
KNN	0.990	0.993	0.991
LOF	0.906	0.698	0.789
One-Class SVM	0.383	0.899	0.537
Isolation Forest	0.403	0.697	0.510
Elliptic Envelope	0.435	0.800	0.564
PCA	0.370	0.949	0.532
GMM	0.739	0.799	0.768
K-Means	0.258	0.141	0.182
DBSCAN	0.324	0.555	0.409
Autoencoder	0.383	0.945	0.545
AdaBoost	0.825	0.756	0.789

Combined Machine-Generated Text Benchmark

Chapter 6 reports the LLM-embedding setting for the pooled five-dataset benchmark. The appendix adds the remaining pooled baselines.

Table A.13: Combined machine-generated text appendix-only results using the mixed-data baseline.

Method	Precision	Recall	F1-score
KNN	0.475	0.344	0.399
LOF	0.679	0.491	0.570
One-Class SVM	0.714	0.517	0.600
Isolation Forest	0.501	0.363	0.421
Elliptic Envelope	0.829	0.600	0.696
PCA	0.492	0.356	0.413
GMM	0.792	0.573	0.665
K-Means	0.568	0.411	0.477
DBSCAN	0.553	0.400	0.464
Autoencoder	0.681	0.493	0.572
AdaBoost	0.923	0.958	0.940

Table A.14: Combined machine-generated text appendix-only results using the transformer-embedding pipeline.

Method	Precision	Recall	F1-score
KNN	0.963	0.950	0.956
LOF	0.598	0.689	0.640
One-Class SVM	0.544	0.902	0.679
Isolation Forest	0.469	0.701	0.562
Elliptic Envelope	0.522	0.803	0.633
PCA	0.431	0.951	0.593
GMM	0.654	0.882	0.751
K-Means	0.036	0.032	0.034
DBSCAN	0.446	0.471	0.458
Autoencoder	0.471	0.950	0.630
AdaBoost	0.826	0.786	0.805

A.2 AUDIO DATA

MIMII

Chapter 6 uses the log-Mel/MFCC statistics. The appendix adds the omitted aggregate embedding-based results.

Table A.15: MIMII appendix results using the embedding-based pipeline.

Method	Precision	Recall	F1-score
KNN	0.263	0.749	0.379
LOF	0.279	0.798	0.402
One-Class SVM	0.206	0.604	0.299
Isolation Forest	0.205	0.594	0.296
Elliptic Envelope	0.224	0.635	0.323
PCA	0.249	0.711	0.359
GMM	0.258	0.725	0.370
K-Means	0.224	0.638	0.322
DBSCAN	0.177	1.000	0.295
Autoencoder	0.267	0.763	0.385
AdaBoost	0.317	0.903	0.456

ASVspoof 2019

The main chapter keeps the log-Mel/MFCC statistics. The table below reports the omitted wav2vec-based results.

Table A.16: ASVspoof 2019 appendix results using the wav2vec feature pipeline.

Method	Precision	Recall	F1-score
KNN	0.915	0.103	0.185
LOF	0.940	0.126	0.223
One-Class SVM	0.930	0.292	0.444
Isolation Forest	0.942	0.195	0.324
Elliptic Envelope	0.906	0.027	0.052
PCA	0.860	0.573	0.688
GMM	0.944	0.192	0.319
K-Means	0.934	0.136	0.237
DBSCAN	0.952	0.045	0.087
Autoencoder	0.848	0.292	0.435
AdaBoost	0.886	0.984	0.932

A.3 VIDEO DATA

UCSD Anomaly Detection Dataset

Chapter 6 already reports the ResNet-based frame-level comparison. To keep this appendix non-duplicative, the tables below retain the omitted HOG and optical-flow variants, two additional Vision Transformer evaluations, the memory-augmented autoencoder experiment, and the extra convolutional-autoencoder row that was not included in the common Chapter 6 table.

Table A.17: UCSD results using the memory-augmented autoencoder pipeline.

Method	Precision	Recall	F1-score
KNN	0.353	0.103	0.159
LOF	0.133	0.039	0.060
One-Class SVM	0.392	0.114	0.177
Isolation Forest	0.322	0.094	0.145
Elliptic Envelope	0.322	0.094	0.145
PCA	0.392	0.114	0.177
GMM	0.467	0.136	0.211
K-Means	0.400	0.117	0.181
DBSCAN	0.314	0.091	0.142
AdaBoost	0.378	0.110	0.171
MemAE	0.433	0.126	0.196

Table A.18: UCSD results using the Vision Transformer evaluation.

Method	Precision	Recall	F1-score
KNN	0.436	0.127	0.197
LOF	0.225	0.066	0.102
One-Class SVM	0.458	0.134	0.207
Isolation Forest	0.394	0.115	0.178
Elliptic Envelope	0.453	0.132	0.204
PCA	0.478	0.139	0.216
GMM	0.517	0.151	0.233
K-Means	0.456	0.133	0.206
DBSCAN	0.381	0.111	0.172
AdaBoost	0.436	0.127	0.197
MemAE	0.467	0.136	0.211

UCF-Crime

Chapter 6 reports the 3D-convolutional setting. The omitted Vision Transformer comparison is retained below.

Table A.19: UCF-Crime results using the Vision Transformer feature pipeline.

Method	Precision	Recall	F1-score
KNN	0.701	0.128	0.217
LOF	0.731	0.134	0.226
One-Class SVM	0.719	0.132	0.223
Isolation Forest	0.731	0.134	0.226
Elliptic Envelope	0.766	0.141	0.237
PCA	0.737	0.135	0.228
GMM	0.808	0.148	0.251
K-Means	0.743	0.136	0.230
Autoencoder	0.737	0.135	0.228

WildDeepFake and Celeb-DF

The main chapter keeps the 3D-convolutional setting for WildDeepFake and the Vision Transformer setting for Celeb-DF. The appendix therefore records the remaining video representations.

Table A.20: WildDeepFake results using the CNN feature pipeline.

Method	Precision	Recall	F1-score
KNN	0.593	0.540	0.565
LOF	0.529	0.660	0.587
One-Class SVM	0.484	0.060	0.107
Isolation Forest	0.479	0.364	0.414
Elliptic Envelope	0.490	0.350	0.408
PCA	0.500	0.032	0.060
GMM	0.571	0.024	0.046
K-Means	0.529	0.276	0.363
DBSCAN	0.508	1.000	0.674
Autoencoder	0.480	0.320	0.384
AdaBoost	0.676	0.560	0.613

Table A.21: WildDeepFake results using the Vision Transformer feature pipeline.

Method	Precision	Recall	F1-score
KNN	0.590	0.490	0.535
LOF	0.515	0.709	0.597
One-Class SVM	0.354	0.039	0.071
Isolation Forest	0.478	0.300	0.369
Elliptic Envelope	0.518	0.342	0.412
PCA	0.593	0.062	0.112
GMM	0.620	0.048	0.088
K-Means	0.645	0.257	0.368
DBSCAN	0.555	0.999	0.713
Autoencoder	0.522	0.584	0.552
AdaBoost	0.685	0.457	0.548

Table A.22: Celeb-DF results using the CNN feature pipeline.

Method	Precision	Recall	F1-score
KNN	0.875	0.995	0.931
LOF	0.882	0.457	0.603
One-Class SVM	0.923	0.186	0.309
Isolation Forest	0.885	0.442	0.589
Elliptic Envelope	0.879	0.459	0.603
PCA	0.940	0.099	0.180
GMM	0.917	0.062	0.117
K-Means	0.863	1.000	0.927
DBSCAN	0.863	1.000	0.927
Autoencoder	0.665	0.272	0.386
AdaBoost	0.863	1.000	0.927

Table A.23: Celeb-DF results using the 3D-convolutional feature pipeline.

Method	Precision	Recall	F1-score
KNN	0.869	0.992	0.926
LOF	0.850	0.353	0.499
One-Class SVM	0.798	0.053	0.100
Isolation Forest	0.824	0.286	0.424
Elliptic Envelope	0.822	0.272	0.408
PCA	0.875	0.050	0.094
GMM	0.645	0.014	0.028
K-Means	0.863	1.000	0.927
DBSCAN	0.863	1.000	0.927
Autoencoder	0.860	0.026	0.051
AdaBoost	0.863	1.000	0.927

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